

Engineering Efficient Self-Play Chess: Search, Replay, and Throughput Under Limited Compute

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Abstract

How strong can an AlphaZero-style chess system become under limited training compute when its entire learning loop is engineered for efficiency? We train from random initialization through searched self-play on a single eight-GPU node for 2.5 days. The resulting 6.32-million-parameter model reaches 3,251 benchmark Elo [3,206, 3,297] at 100,000 searches per move against a fixed-node Stockfish 13 ladder. The run ingests 3.25 million completed games, involves an estimated 195 billion search-time neural-network evaluations, and makes 836.6 million training presentations. We investigate search allocation, replay and restart-state selection, policy representation, progressive model sizing, quantized inference, and throughput engineering. Alongside the retained design, we document plausible alternatives that failed to improve the complete learning loop or did not justify their cost. The reported strength is a result of the integrated system, not an isolated Elo gain attributable to any single choice.

Keywords: AlphaZero, chess, self-play, Monte Carlo tree search, replay, GPU inference, quantization, model growth, fixed-node evaluation

1 How far can efficient self-play go?

An AlphaZero-style chess player [1] improves by searching its own games, learning from the resulting positions, and repeating that cycle with a stronger network. Search creates targets; replay decides which targets persist; training absorbs them; and evaluation must distinguish progress from noise. Under a limited budget, improving any one step matters only if the complete loop produces stronger play within the available time.

This study asks how strong that loop can become on a single eight-GPU node when the system is engineered for efficiency. Over **2.5 days** of training from random weights, using searched self-play rather than human-game training targets, the run produced a 6.32-million-parameter model. It reached **1,658 benchmark Elo without search** and **3,251 benchmark Elo at 100,000 searches per move** against the project’s fixed-node Stockfish 13 ladder. A matched-estimator comparison places it about **74 Elo** above the previous training baseline at a 64-search plateau. These are results under specific match protocols, not FIDE ratings or unrestricted-engine rankings. The terminal games and intervals appear in Chapter 8 and Appendix B.

The central lesson is that the choices interact. Search determines which targets are worth paying for; replay determines which targets the learner sees again; and representation affects both learning and inference cost. Native search, batched inference, and TensorRT provide the throughput that makes those choices consequential. Several plausible optimizations improved a local proxy without improving the complete learning loop.

1.1 Scope and contributions

Chess is the subject of the study. The runtime also supports 7×7 and 9×9 Go, which helped validate the shared platform and supplied ideas such as KataGo’s fast and full searches [7]. Small-board Go was briefly considered as a cheaper setting for tuning chess hyperparameters. Its first-player advantage, shorter games, rapidly learned value target, and apparent need

for different tuning made that transfer unattractive. This is the project owner’s qualitative rationale, not a controlled cross-game finding; the report makes no Go strength claim.

The report presents the training recipe and terminal evaluation, then examines the choices that shaped them: search allocation, graph search and caching, policy representation, model sizing, replay and restart states, resignation, auxiliary targets, and quantized inference. It also explains the throughput path needed to supply searched games and studies three failures with transferable lessons about self-play targets, deployment fidelity, and model promotion. The readable, living recipe begins at `chess-final-config.yaml` [10]; the reported checkpoint is identified separately from that configuration.

The final recipe combines many changes. Its strength establishes the outcome of the assembled system, not an isolated Elo contribution for each component. We use paired games for playing-strength claims, keep throughput and target-fidelity measurements tied to their own protocols, and distinguish preserved evidence from owner recollection when historical result artifacts are missing.

A live chess demonstration is available [12], but its games are not part of the evaluation protocol.

1.2 Reading the report

Chapter 2 defines how evidence is interpreted; Chapter 3 presents the learning loop. The investigation chapters then ask how to spend search, what the network predicts, and which positions become training data. A short systems chapter explains how enough games were produced. Three failure studies lead into the integrated recipe and final evaluation. Implementation records are available in the project repository [10], but the paper states the methods and results needed for its own conclusions.

2 How claims are measured

The evidence required depends on the claim. Faster inference establishes a local throughput gain, and lower policy

loss shows better fit to a given target. Neither demonstrates stronger play. In online self-play, concurrent training can absorb saved search time, while a network that fits stored positions may produce a worse distribution of new games. We therefore separate playing strength, learning progress, throughput, and numerical fidelity throughout the report.

2.1 Playing strength

The terminal chess evaluation plays the selected checkpoint against Stockfish 13 at fixed node limits. Openings are paired: each starting position is played once from each colour. The protocol fixes the opening suite, opponent version and resources, candidate artifact, search budget, and parallelism; Appendix B reports the node limit, W/D/L count, score, and interval for each terminal match row.

The opponent nodes are assigned historical ratings from Marco Meloni’s fixed-node Stockfish calibration [9]. The result is **benchmark Elo** under this protocol, not a FIDE or online rating or a ranking against unrestricted contemporary engines. At each model search budget, the headline uses the tested opponent whose score is nearest 0.5, limiting extrapolation; both tested limits remain visible. Their implied ratings disagree most at shallow model budgets, and the available matches do not isolate why. The reported bootstrap intervals quantify match sampling while holding the calibration anchors fixed, so they do not include calibration uncertainty.

The policy-only row selects legal moves directly from a float TorchScript policy export. Searched rows use the INT8 TensorRT deployment artifact. This distinction is part of the result, not an assumption of backend equivalence. Search parallelism also changes along the deep-search curve; the curve describes achieved operating points, not a controlled fixed-parallelism scaling law.

2.2 Learning and comparison

To compare online learning interventions, we measure playing strength over time while matching initial weights, replay contents, evaluation, and hardware workload as closely as the experiment permits. Independent from-scratch trajectories can vary more than a small proposed effect; the decisive adaptive-search comparisons therefore started from the same model and replay state. Frozen-replay fits screen optimizers, policy heads, and quantization schemes, but cannot establish online self-play strength by themselves.

The cross-campaign 64-search plot is descriptive. The previous baseline and final recipe used different ladder estimators, so subtracting their plotted endpoints would not be a valid strength comparison. A retrospective three-rung comparison gives an approximately 74-Elo plateau difference, subject to an approximately ± 15 -Elo estimator-transfer sensitivity. That sensitivity is not a game-level confidence interval; Appendix B gives the comparison boundary and transfer assumption.

2.3 Throughput and target fidelity

Throughput has several distinct units: model evaluations, search simulations, completed games, materialized replay positions, optimizer steps, and strength gained per wall-clock hour. A gain at one stage may vanish at the next. Rate claims therefore specify the relevant batch size, concurrent games, GPU contention, search parallelism, and training overlap. Sat-

urated many-game throughput is not interactive single-game latency.

Serving artifacts must be checked as chess models, not merely as files that load. The TensorRT refit investigation showed that successful export and refit calls could still yield incorrect legal-move probabilities. Fidelity checks therefore compare outputs on real encoded positions, including legal-action masking, policy agreement and Kullback–Leibler divergence, and value error. A float checkpoint’s match cannot substitute for a match by the INT8 artifact intended for deployment.

2.4 Provenance and limits

The living final configuration [10] explains the current recipe; it is not by itself a frozen record of the run reported here. Reproduction also requires the source revision, resolved configuration, selected checkpoint and deployment artifacts, evaluation assets, and archived results. Appendix D distinguishes public artifacts from locally archived evidence. A proposal or configuration records intent; implemented code and preserved measurements are needed to establish that an experiment happened.

Many choices in the final recipe changed together. We call a choice *retained* when it appears there, and attribute an isolated effect only where a controlled comparison supports one. Missing historical policy-head results and qualitative owner recollections are identified rather than converted into measurements. The methods and results needed for each conclusion are stated in the paper; project records provide provenance, not missing premises.

3 System and training method

3.1 Runtime boundary

The implementation matters to the research question because the learner needs enough searched games and fresh positions within the reported 2.5-day run. Figure 1 follows that loop from model publication through self-play, replay, training, and evaluation. This chapter describes the interfaces that determine its throughput and targets; Chapter 5 examines their measured costs.

Native C++ owns game state, legal actions, encodings, tree search, self-play, and batched inference. Python coordinates validated configuration, workers, replay, distributed training, model publication, and evaluation. There is no second production rules or search implementation in Python. Tests enforce the cross-language action mapping, tensor shapes, feature planes, symmetry, and output order.

3.2 Chess representation and outputs

The network receives a board-aligned tensor of current state, history, and additional chess features. Its reduced action space has 1,880 chess actions; a game symmetry transforms the position and its action targets together. The serving artifact exposes only policy and win/draw/loss value outputs. Auxiliary heads train the shared representation but are absent from inference.

The retained network is convolutional, with scaled post-activation residual blocks, capped activations, and global-pooling context in every second block. A from-to attention policy head scores move origins and destinations more com-

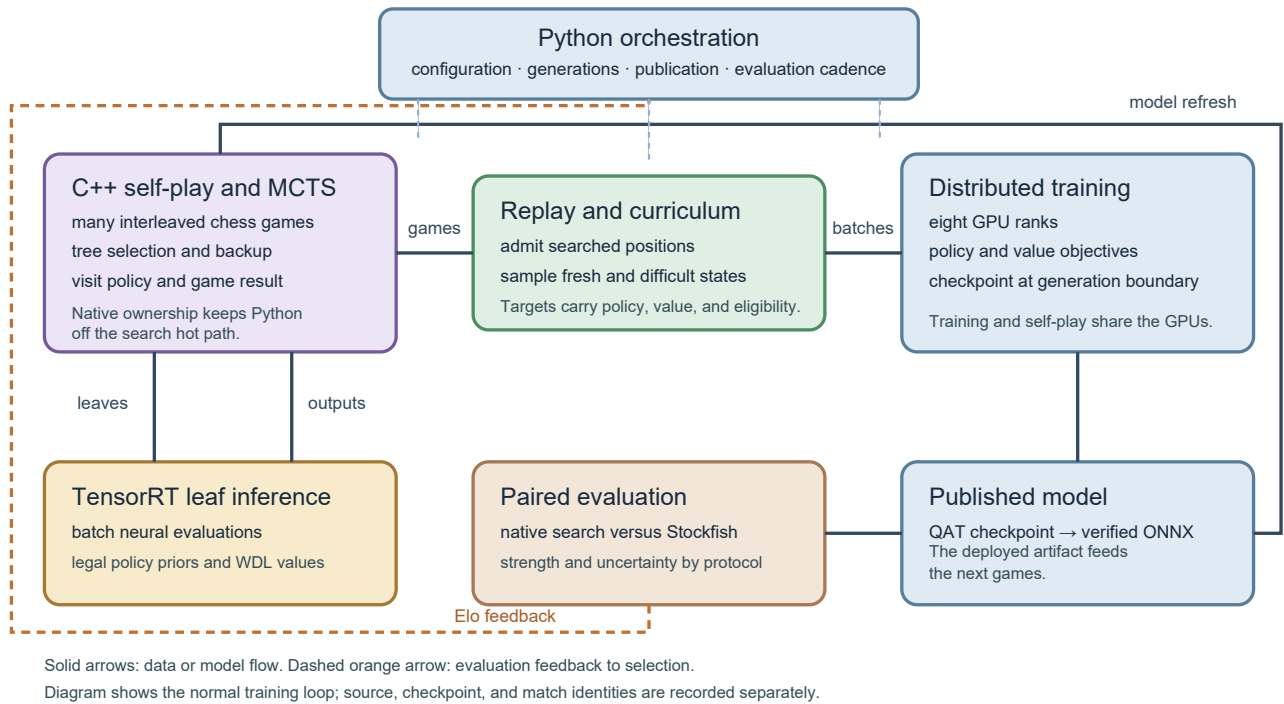


Figure 1: A published model returns to batched native self-play; searched positions become replay targets for the trainer. Paired evaluation measures the published checkpoint and feeds selection decisions back to Python. The diagram shows ownership and data flow, not the number or scheduling of every worker.

pactly than a dense action projection. The value path reduces to two channels before a small fully connected layer. Chapter 4 examines the representation alternatives; Chapter 7 gives the selected shapes.

3.3 Search and self-play

Self-play uses PUCT-style Monte Carlo tree search with neural policy priors and value estimates. Search visits follow a staged fixed budget, rather than a budget adapted separately to each position. Root exploration uses Dirichlet noise, reduced-parent-value first-play urgency, and forced playouts; a fraction of root visits is retained after a move. Chapter 4 explains why the tested adaptive alternatives were not retained.

Thousands of games are interleaved to fill GPU batches. Native workers retain their trees; one inference worker per process submits batches to TensorRT. Parallel simulations improve batch fill but can select leaves using stale search information. Concurrent games can raise aggregate throughput while lengthening an individual game. The chosen search and topology therefore form one operating point, not independent speed knobs.

Starting positions mix shallow random legal openings with archived restart states. Restart candidates favor uncertain, consequential positions rather than uniformly sampling history. The random-opening share preserves games that evolve from the opening, while restarts revisit positions where additional search may matter. Resignation is enabled only after a calibrated false-nonloss gate is met; continuation games check the gate and retain terminal examples.

3.4 Replay and materialization

Native workers publish complete trajectories atomically. Parallel materializers reconstruct observations into fixed-layout

columnar shards; a circular memory-mapped store is the training replay. Search policies remain sparse in storage and are densified only when a training batch is formed.

Each row retains the state, legal moves, search policy, WDL target, root value, sample weight, source generation, policy-surprise information, and configured auxiliary targets. Exactly-once trajectory claims and rejection telemetry protect the boundary between completed games and admitted replay rows.

Sampling mixes uniform draws with policy surprise: positions where search disagreed with the network prior are drawn more often without losing broad coverage. Replay capacity grows in stages. A configured replay ratio ties optimizer work to materialized positions, preventing training from silently outrunning data generation. Chapter 4 separates the measured effects of these data choices from the integrated recipe.

3.5 Training

Eight persistent NCCL trainer ranks, one per GPU, use a global batch of 2,048 in bfloat16. Training runs in 500-step optimizer blocks while some self-play workers remain active. Each training-and-publication cycle is called a *generation*; a *checkpoint* is its durable model artifact. Because training and self-play overlap, fewer search simulations do not necessarily shorten wall-clock training. This distinction is central to the adaptive-stopping result in Chapter 4.

The primary objective combines policy cross-entropy with WDL/value loss. Outcome value is discounted by ply, and a small scheduled blend of search-root value is introduced later. Two training-only targets—the next position’s search policy and normalized remaining game length—provide auxiliary learning signals. Search does not query them.

3.6 Progressive models and publication

Training begins with a smaller network because its higher inference throughput supplies more early games. The production controller trains the next larger candidate from an independent initialization alongside the active model. A strength-improvement plateau triggers candidate training, while two qualifying head-to-head matches gate promotion. Function-preserving growth from a trained medium model was tested separately, not as the controller’s ordinary path. Chapter 7 gives the selected schedule and gate.

Rank zero publishes both a recoverable training checkpoint and a trimmed inference artifact. The final path refits quantization-aware-trained ONNX weights into TensorRT templates for self-play and evaluation. Numerical fidelity is checked at publication; Chapter 6 describes the failure that made this check necessary.

3.7 Evaluation

Short-lived evaluations run on a fixed cadence beside training. Fixed-dataset metrics and paired-opening Stockfish ladders monitor policy-only and searched play. Terminal evaluation uses larger matches and deeper search budgets. The run records the engine binaries, datasets, opening books, and their hashes as part of the protocol.

The training ladder brackets the candidate against nearby Stockfish node limits instead of extrapolating from one opponent. Paired openings reverse colors, and reports retain W/D/L and search identity. Batch shape can alter serving behavior, so it is part of the evaluation protocol. Frequent ladders provide a noisy progress signal; the terminal paired matches in Appendix B support the final strength estimate.

Completed trajectories, replay state, checkpoints, and evaluation evidence are committed before they are credited or reported. Appendix D states which artifacts are public and which remain in the local archive.

4 Research investigations

4.1 Spending search where it matters

Search made a fixed network substantially stronger: in one frozen-model comparison, increasing the budget from 200 to 1,600 visits raised its score against the same opponent from 0.318 to 0.748. The harder question was whether those visits could be spent more usefully during training. A cheaper move might complete a game sooner, but it may also create a weaker policy target or leave the accelerator underfilled. None of the tested adaptive allocators improved the end-to-end learning loop, and exact graph or cache reuse did not repay its overhead. The retained answer is a staged, fixed visit cap for every recorded move.

That conclusion depends on what is measured. Fixed-network playing strength tests the moves chosen by search; target fidelity compares a shallow visit distribution with a deeper reference; online learning measures the next network; and throughput measures work completed per unit time. An improvement on one axis did not necessarily transfer to another. Chapter 2 defines the measurement conventions.

4.1.1 The fixed-budget baseline

The native engine performs policy-guided Monte Carlo tree search. The network supplies legal-move priors and a win/draw/loss value; PUCT selects leaves; neural evaluations are batched across active games; and values are backed up from the alternating player’s perspective. A visit cap is deliberately unambiguous. It does not depend on a difficulty estimate, a calibrated controller, or a learned model that may become stale as training advances.

Against the same opponent, score rose from 0.318 at 200 visits to 0.537, 0.580, 0.662, and 0.748 at 400, 600, 1,000, and 1,600 visits. Across the better-resolved middle and deep part of the sweep, the fitted trend was about 81 Elo per doubling. Individual adjacent comparisons were noisy, so the overall trend is more credible than any single increment. Search was still changing its policy target too: at 600 visits, only 72.4% of positions selected the same best move as the network’s own 10,000-visit reference. These frozen-network measurements establish the value of search depth, not an optimal training schedule.

The retained schedule therefore increased the cap in auditable stages—300, 400, 500, 600, and finally 800 visits—rather than presenting those boundaries as independently optimized discoveries. Even this count needs care: a retained root can begin a move with existing statistics, so a nominal visit cap is not always the amount of new work performed.

4.1.2 Why fast and full searches did not transfer

A scheme inspired by KataGo’s self-play methods [7] used cheap searches to advance most moves and full searches on a random minority, which alone became primary policy targets. The trade is attractive in 19×19 Go: long games make finished outcomes expensive, so cheap moves can supply more independent value targets. Chess games are shorter, their outcomes and positions are often clearer, and the value objective was already learning. The transferred recipe thus spent search on moves whose positions were discarded as primary training rows without showing that the extra completed games compensated for the lost policy-target density.

The cheap moves were not free. With a quarter of moves searched to 600 visits and the rest to 150, the cheap moves still consumed 42.9% of nominal search work. Nor were they isolated from training: they selected played moves, affected terminal outcomes and retained trees, could supervise a preceding row’s next-policy target, and influenced which positions entered the restart archive. “Not stored as a primary policy row” was never the same as “irrelevant.”

The mixed workload also interacted badly with batching. Once the cheap searches finished, only the full-search minority remained. In a 512-game test this tail left about 128 active trees and filled only 86 of 320 available batch slots with one leaf per tree. Allowing four leaves per tree raised the average batch to about 268 and improved search throughput by roughly 20%, but it did so by making each search less serial. A policy intended to save compute thus created pressure to accept a search-quality tradeoff merely to keep the accelerator occupied.

Most importantly, a forced cheap-search tail made the endgame failure self-reinforcing. It removed searched endgame rows because cheap moves were not admitted as pri-

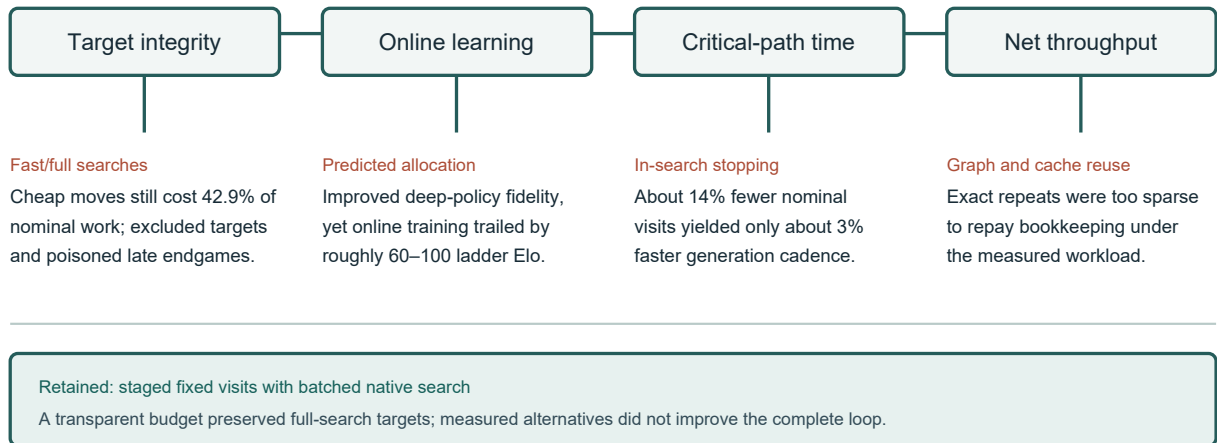


Figure 2: Each rejected idea met a different constraint. Fast/full search damaged target coverage and endgame semantics; predicted allocation improved its fidelity proxy but lost online strength; in-search stopping saved mostly non-critical-path work; and exact graph or cache reuse failed to repay overhead. These are results under the measured chess workload, not universal impossibility claims.

many targets. A shallow estimate at the ply cap could then be propagated through many earlier rows of the same game. The original motive was reasonable—avoid filling replay with noisy positions from very long endings—but it also withheld the examples needed to learn conversion. One full search at the cut position replaced the heuristic bootstrap; later, removing the cheap tail restored searched endgame positions. Because multiple changes accompanied the observed recovery, the record does not isolate either repair as its sole cause (Chapter 6).

The retained design keeps KataGo’s useful distinction between search cost and target eligibility, but searches every recorded move to the scheduled cap. This is a systems-and-target-semantics decision, not a matched-compute Elo proof that all-full search always wins; no such long-run ablation was preserved.

4.1.3 Three attempts to allocate search adaptively

The first attempt asked whether search could stop when the visit leader looked uncatchable. An audit of completed trees found apparent room: an optimistic reconstruction suggested that a rule requiring at least 75% of the cheap budget might remove 15.3% of cheap-search visits, or 6.38% of all nominal limits if full searches saved nothing. But the records contained final visit distributions, not the intermediate traces needed to identify when the leader first became safe. They also omitted enough state to reconstruct retained visits and forced-playout-pruned targets. The apparent saving was therefore an oracle-like opportunity estimate, not an observed stopping result.

A wider offline study found a deeper problem. The marginal value of search was not monotone in visit concentration: very diffuse and already-decided positions gained little, while moderately concentrated but contested positions gained most. Simple concentration thresholds tended to stop in the very region where more search was useful. The rule-based approach was audited and declined before an online strength match; it should not be described as an implemented algorithm that lost Elo.

The next allocator replaced the rule with a learned predic-

tion, related in aim but not identical to dynamic simulation stopping [3]. An auxiliary head estimated, for several candidate budgets, the divergence between that budget’s policy and a deep-search policy. A calibrated corrector incorporated root observables, and a dual variable kept average spend near its target. Deep labels, replay persistence, model publication, native budget selection, safety gates, and telemetry were all implemented. Mechanically, the system worked. At approximately matched mean spend it captured about 23% of the available policy-divergence headroom; its mean target fidelity resembled roughly 1.18 times uniform search at 0.967 times the spend. It also behaved plausibly, assigning more work to contested positions.

Learning nevertheless deteriorated. Repeated online attempts trailed comparable non-adaptive training by roughly 60–100 ladder Elo. More than a third of positions received an average budget fraction near 0.36, and almost 9% received one eighth of baseline search. Those shallow policies were close to the network’s own prior yet were trained at full weight. The likely failure was therefore objective mismatch: closeness to a deep policy at the current position does not measure how much a target will improve the next network. That mediation remains a hypothesis, but the central result does not: the controller improved its stated proxy and still produced worse learning.

The final adaptive system moved the decision inside search, where a learned stopper could observe the evolving tree rather than predict difficulty in advance. Its decisive test started from the same checkpoint, optimizer, and rebuilt replay state for every arm. The most aggressive setting skipped about 14% of nominal search, and its internal credit-wait measurements changed in the expected direction. Yet generation cadence improved by only about 3%. Self-play overlapped the optimizer, so most of the removed search was slack rather than critical-path work. Paired strength estimates were $+1.7 \pm 9.9$ Elo and -4.2 ± 10.1 Elo (standard errors) for the two stopping settings: neither resolved a strength effect. At the observed learning rate, the cadence gain was worth only about one Elo over three hours, below the experiment’s resolution.

The failures have different boundaries: the threshold rule

lacked an identifiable safe signal; the predicted allocator improved policy fidelity but not learning; and the in-search stopper removed mostly off-critical-path work. A non-overlapped or inference-bound learner could change the last result. None establishes that adaptive search is universally ineffective.

4.1.4 Parallel leaves: buying latency with search quality

Batching across many independent games is the cleanest way to feed the accelerator, but the number of active roots eventually runs out. The engine can then keep several leaf traversals in flight from one root. Virtual reservations discourage those traversals from selecting the same path, yet every selection is based on a tree that is missing the other in-flight results. Parallel search is therefore not serial MCTS executed faster. It exchanges fresher decisions for larger batches and lower latency.

The experiments exposed how easy it is to measure the wrong regime. An initial sweep suggested little cost per doubling, but its batch capacity was too small relative to the number of trees, so the configured per-root parallelism barely bound. A deliberately oversized-batch rerun activated the knob and estimated a steeper -6.4 ± 4.7 Elo per doubling. The interval still included a small effect, and the rerun was diagnostic rather than a production throughput configuration.

The tradeoff also depended on the total budget. At 1,000 searches, a terminal fixed-network sweep measured point losses of 19 Elo with four leaves and 45 Elo with sixteen leaves relative to serial search, while sixteen-way parallelism was substantially more damaging at 100 searches. The likely explanation is that a deep search will eventually visit more of the temporarily suboptimal leaves selected from stale state. However, only a few budget and parallelism combinations were measured, and the terminal strength curve itself uses different parallel counts at different depths. It is not a pure scaling curve from which a universal safe-parallelism law can be inferred.

The retained design consequently uses bounded parallelism as a serving parameter, not a free algorithmic speedup. The right setting depends on available independent roots, batch capacity, total search depth, latency requirements, CPU load, and memory. Older results from the fast/full batching tail do not directly prescribe the topology of the final all-full workload.

4.1.5 When a tree became a graph

Chess appears rich in transpositions: different move orders often reach the same board. The project tested whether Monte Carlo graph search, as in Czech, Korus, and Kersting [6], could turn those transpositions into shared neural evaluations, descendants, and search statistics. This was a complete implementation, not a cache mislabeled as graph search. Canonical nodes held shared state-level information; parent/action edges retained local PUCT statistics; correction backups exposed better shared values to incoming edges; and the system handled trajectory reservations, cycles, rerooting, pruning, and capacity reclamation. An audit against both the paper and its reference implementation found and corrected a missing first-link backup behavior.

The limiting fact was chess-state identity. Pieces and side to move are not enough: castling rights, en-passant state, the

halfmove clock, and repetition-relevant history can change the legal result. Merging positions that differ on those fields would create an approximate algorithm with different game semantics. Under exact equality, most apparent board transpositions disappeared.

At ordinary budgets, verified links were absent or negligible. After the first-link correction, only 0.0249% and 0.1769% of neural evaluations were avoided at 1,000 and 10,000 searches, while the graph was 8.63% and 8.28% slower. At 30,000 and 60,000 searches, table hits rose to 2.37% and 3.46%, but avoided evaluations remained approximately 0.0001% and 0.0348%; throughput was still 7.06% and 5.76% lower. Structural counters confirmed that shared descendants and statistics were active. A final strength match was unnecessary to answer the deployment question: exact reuse was orders of magnitude too sparse to repay the book-keeping cost. This is an implementation and throughput rejection, not evidence that graph search is universally ineffective.

4.1.6 Why inference caching found little to reuse

Inference caching asked a narrower question: if an identical encoded neural input reappears under the same model, can its policy and value outputs be reused without sharing search state? Two investigations answered complementary parts of it.

First, a real bounded, sharded cache was shared by the search threads within each self-play process. With eight processes per GPU, three search threads per process, and capacity for 1.5 million entries per process, its hit rate was 0.970%. Disabling the cache made game updates about 0.88% faster in the short stochastic comparison and reduced summed worker peak memory by about 7.12%. The implemented cache consumed memory without demonstrating a speedup.

A later audit measured the upper bound available to a wider cache before building one. An unbounded tracker shared across one search executor observed exact encoded inputs but deliberately evaluated every position. With diverse starts and production-style progression, repeats were 1.33% at 150 searches, about 4.24% at 800, and about 3.5% in the mixed workload; same-batch duplicates were essentially absent. Following retained trees for several moves raised the rate only to roughly 4%. The tracker itself cost about 3.65% throughput and grew without bound. A real cache would additionally pay for output storage, synchronization, eviction, and device transfers while retaining fewer entries.

The wider design was declined before implementation; it was an opportunity audit, not a production cache benchmark. Together with the implemented per-process cache, it bounds the likely benefit under this workload.

4.1.7 The retained search design

The selected system returned to fixed visits, but not to a bare or naive search. It uses PUCT, reduced-parent-value first-play urgency, root noise and temperature for self-play exploration, forced root playouts followed by pruning of visits that should not become policy supervision, batched native inference, a 0.99 per-ply value discount, and tree retention with 60% of visits carried across a played move. Every recorded move receives the current staged visit cap. The current settings are

summarized in Chapter 7 and defined by the public configuration [10].

Those ingredients do not form an additive ablation table. Search depth has direct strength and target-fidelity evidence. Evaluation and self-play clearly need consistent first-play-urgency semantics. Parallel leaves have a measured, workload-dependent quality/latency tradeoff. Root retention, forced playouts, noise, discount, and the exact PUCT constants are retained parts of a coherent AlphaZero-style recipe, but most lack isolated final-workload strength estimates.

Under the measured chess workload, fixed visits made target provenance and search expenditure explicit while the tested alternatives failed to show a better learning return. A repetitive analysis service or a non-overlapped learner could change the economics, but that possibility does not substitute for a measured improvement here.

4.2 Getting more learning from each game

More games help only if they add positions with interpretable targets. The retained chess curriculum diversifies starts, stores searched positions in a growing replay window, samples policy-surprising rows more often, and funds four training presentations per admitted row. It also distinguishes a resolved outcome from a game stopped at the ply cap. The clearest direct target comparison favored a search-root bootstrap over a material heuristic at that cap; most other curriculum choices were retained together and have no isolated Elo estimate. That distinction between measured target quality and an integrated recipe governs this chapter.

The project separated four operations that are easy to confuse. *Generation* chooses the trajectories to search. *Admission* turns eligible observations into durable rows. *Selection* chooses rows for a batch. *Weighting* changes their contribution after selection. A fifth mechanism, presentation credit, governs when training may advance. This vocabulary matters because a technique that prioritizes future games is not the same as prioritized replay, and drawing a row more often is not the same as increasing its loss weight.

4.2.1 Replay is both memory and clock

A replay window mediates between freshness and breadth. A small FIFO quickly removes targets made by weak old policies, but concentrates learning on a narrow recent distribution. A large window preserves more openings, endgames, and policy eras, yet can keep stale targets alive after the acting model has moved on. Allocating room for twenty million rows does not itself create twenty million distinct positions.

The retained design therefore preallocates the physical memory map once while growing its logical capacity through 0.6, 1.2, 2.0, 2.8, 4, 6, 8, 12, 16, and 20 million rows. Growth avoids pretending that an empty early window offers diversity and later allows a broader policy history to remain available. A mature earlier campaign showed why freshness must be measured rather than inferred: strength continued to improve after fixed-dataset policy accuracy largely saturated, while larger models and deeper search reduced the rate of new positions. That observation motivates the wider window but does not isolate this exact schedule as a strength improvement.

Replay reuse introduces a second tradeoff. The configured ratio is the number of optimizer presentations funded by each

newly admitted row. In the retained setting, four presentations are credited per row; a 500-step quantum at a global batch of 2,048 therefore requires 256,000 newly appended rows. Credits are issued only after append and flush, and a persistent ledger prevents a restart from spending them twice.

This ratio is not merely a sample-efficiency knob. Because training, evaluation, model publication, visit schedules, and replay growth advance at quantum boundaries, changing reuse also changes their wall-clock pace. Higher reuse funds more optimizer work from the same generated data; lower reuse exposes each update to more fresh positions if the actors can supply them. Short controls at ratios four, 6.25, and eight remained matched at their shared evaluation boundaries despite different update rates, but all ended within roughly 90 minutes and their full raw curves are not preserved in the benchmark archive. The completed campaign also changed replay capacity, optimizer, objective weighting, and inference alongside reuse. Ratio four is consequently a deliberate freshness bias, not a measured standalone explanation for final strength. Both configured reuse and effective presentations per distinct admitted row are needed to describe what the learner actually saw.

4.2.2 Choosing information without inventing it

Within the live window, the sampler favors rows where search most revised the raw policy. This is related to prioritized replay [4], but uses a different signal and no importance correction. Policy surprise is the divergence between the visit distribution and the network prior. Seventy percent of draws are allocated in proportion to this signal, capped at 2.0, while 30% remain uniform. The cap limits domination by extreme rows and the uniform component preserves broad coverage. The same row may reappear across optimizer steps, but sampling is without replacement within one global batch.

This is prioritization, not an unbiased estimate of the uniform-replay objective. The implementation deliberately does not apply inverse-probability correction, so the optimizer learns from the prioritized distribution. Surprise can also reflect search noise, target age, phase of play, or the number of legal moves—not only genuine difficulty. The sampler is part of the successful retained bundle, but there is no isolated online ablation for its 70/30 mix or cap.

Loss weighting is separate. Each row has a positive sample weight which, after batch-mean normalization, multiplies the primary and eligible auxiliary losses. Rows in the selected recipe use weight 1.0. Earlier replay could aggregate duplicate positions and encode multiplicity through weights; that established the mechanism, not a playing-strength gain from arbitrary weights. Thus the final system sees policy-surprising positions more *often*; it does not increase their per-draw loss weight. TD-error priority, recency weighting, and global deduplication remained proposals.

Admission happens earlier still. Random opening-prefix moves and the reconstructed prefix of a restart game have no search observation and create no replay row. A normal searched move does. A final search performed only to value a cut position can also become a row even though it selects no played action. Sparse policy storage retains at most 60 actions and records how much visit mass was discarded. These rules define the data set before sampling or weighting can act on it.

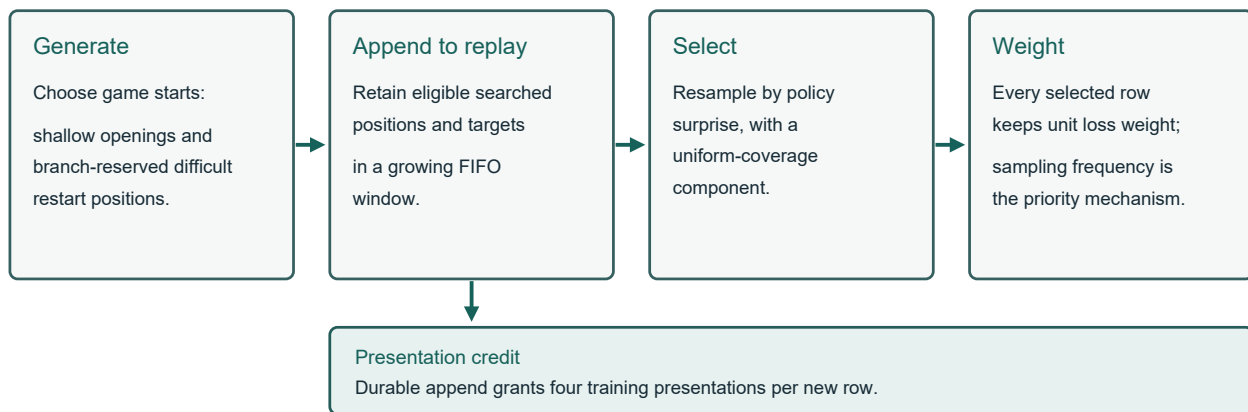


Figure 3: The retained curriculum changes where games start and which valid positions recur, but leaves admitted rows at unit loss weight. The separate presentation-credit ledger allows optimizer work only after replay append and flush; a sampled row is not the same unit as a newly generated row.

4.2.3 Starting games where information is likely

The configured start draw attacks two different coverage problems. Half of games are assigned a uniformly selected zero to eight random legal plies. These prefixes cheaply diversify the opening without pretending to be a balanced opening book. They are reconstructed in the recorded history but not searched or trained directly; their value is the downstream positions they expose. Drawing zero plies naturally retains some games from the ordinary initial position.

The other half are assigned archived self-play states, falling back to random openings when a worker’s restart archive is empty. A position is eligible only if at least 15 plies remained in its source game, its absolute root value was at most 0.8, and two or three leading actions covered 85% of visit mass. The branch actually played is marked used. A later restarted game reserves one untried plausible alternative and forces that action after reconstructing the prefix, so new compute explores a branch the source game did not. This differs from the learned targeted search control [5] explored in prior work: the archive here uses observed search disagreement and untried branches, not a trained restart policy.

Selection from this archive gives 30% probability to uniform choice. The remaining reservations favor the square root of *value correction*: half the absolute difference between the searched root value and the raw network value. The square root softens the priority so one extreme correction cannot dominate. Age and capacity bounds remove old states, reservations prevent duplicate claims within a worker, and exhausted positions leave the archive. This is the project’s practical difficult-state curriculum, but “difficult” is an interpretation of disagreement, not a ground-truth label. The mechanism does not train a regret network, and archives are local to workers rather than globally deduplicated.

Restart priority and replay surprise are related only in motivation. One changes which future trajectories are generated; the other changes which already stored rows are selected for optimization. Their individual Elo effects, the 50% start mixture, and the exact filters were not isolated. They should be presented as a coherent curriculum design grounded in branch coverage rather than as independent measured gains.

4.2.4 Ending games without corrupting their labels

Resignation can save a large amount of search, but a false resignation turns a drawable or winning position into an incorrect terminal label. A fixed value threshold was therefore replaced by continuing measurement. Twenty percent of games are designated at creation as continuation games and can never resign. They reveal the outcomes of hypothetical triggers for thresholds from -0.99 through -0.70. A trigger requires both the root value and the best visited child’s backed-up value to cross the threshold; evidence from capped games is excluded because their natural outcome remains unknown.

Over a rolling window, a candidate threshold needs at least 100 triggered continuation games and a one-sided 95% upper confidence bound on false non-loss no greater than 2.5%. The threshold may become more conservative immediately but relaxes by at most 0.01 per publication. An intentionally aggressive canary verified trigger journaling, persistence, and exclusion of a capped continuation, but did not establish a safe production threshold. The defensible conclusion is that resignation is calibrated and continuously audited—not that it has zero error or a measured independent Elo gain.

A ply cap creates a harder target problem because there is no observed result at all. The project compared material heuristics, raw network value, values from adjacent searches, and a fresh search at the cut position. Playing games beyond the normal cap supplied later outcomes for evaluation. At the earlier measured checkpoint, the cut-position search achieved Brier score 0.444 and cross-entropy 0.756, compared with 0.491 and 0.851 for material divided by 39. At the later checkpoint the corresponding scores were 0.193 and 0.374 versus 0.388 and 0.707. Search-root sign accuracy exceeded 98% in both cohorts; calibration, not merely sign, distinguished the targets. The retained worker therefore performs one full search at the actual cut position and uses its root value as the bootstrap. The scalar root value v becomes a soft WDL target: with $r = 1 - |v|$, the win, draw, and loss components are $\max(v, 0) + r/3$, $r/3$, and $\max(-v, 0) + r/3$. Materialization then applies the configured per-ply blur.

The cut policy addresses the most damaging data failure found in the project. A cheap-search tail excluded late posi-

tions from primary replay while a shallow cutoff estimate supplied the value target for earlier rows in the same capped game. This plausibly reinforced weak conversion: fewer searched endgame examples, more capped games, and more weakly grounded targets. Replay eviction could remove the offending rows without undoing their effect on the network. The measured incidence and multi-change repair are examined in Chapter 6; neither this reconstruction nor the recovery isolates one cause. The general data rule is to audit cut-value provenance and policy-row eligibility together.

4.2.5 Knowing what each target means

Three value mechanisms act at different boundaries and should not be collapsed into “value discount.” During materialization, the terminal WDL target is blurred toward uniform by 0.998 for each remaining game ply. During optimization, a scheduled coefficient eventually blends up to 10% of the stored search-root scalar into that discounted outcome. Inside search, a separate 0.99 per-tree-ply discount changes backups, selection, and the root value before replay is written. The cut-position benchmark supports searched values as better bootstraps than the tested material heuristic; it does not isolate the ordinary 10% blend or either discount as a strength gain.

Future-dependent auxiliary targets require equally explicit eligibility. The retained next-policy target uses the sparse visit distribution from the following searched observation; it is unavailable if that observation does not exist. Remaining game length is known only when the game reaches a natural result or valid resignation, so every row from a cut game marks that target ineligible. The objective masks and renormalizes auxiliary losses over eligible rows rather than treating missing labels as zeros. Overfit tests established that the supplied multi-head targets were learnable, while replay audits established that censoring and offsets were wired correctly. Neither establishes an independent playing-strength gain.

Broader auxiliary bundles were removed during a period with several simultaneous training problems. That was a precautionary simplification, not a controlled negative ablation. Retaining next-policy and remaining length later does not prove that either head helped, just as removing the others does not prove that they harmed. The firm result is semantic: labels derived from the future must be materialized from complete trajectories and masked when the future is unknown.

4.2.6 Fresh targets versus fresh positions

Reanalysis would spend search on fresher targets for existing positions instead of new states and outcomes. An earlier replay design implemented bounded synchronous re-search with source-bound target overrides, but no controlled study compared its learning return with fresh self-play. It disappeared with that replay schema. Reanalysis is therefore superseded infrastructure, not a negative efficacy result.

Model publication addresses freshness from the other side. Publishing every 100 optimizer steps imposed enough serialization, validation, and activation overhead to be rejected, but no preserved matched strength curve gives that choice an effect size. The retained 500-step boundary trades target age against publication cost, and its wall-clock cadence depends on replay reuse.

4.2.7 Overlap without full asynchrony

Training proceeds in coordinated credit-funded quanta while half the actor processes continue self-play. A pause sweep found only a narrow, workload-dependent difference between balanced choices, so the retained fraction is an operational setting rather than a universal optimum. Publication, replay snapshots, and optimizer commits remain synchronized; fully asynchronous learning was proposed but not implemented. The throughput measurements and a corrected device-placement artifact are covered in Chapter 5.

4.2.8 Making the data trustworthy

Completed games enter a typed circular store through bounded, worker-owned materialization. Atomic writes, quarantine with a fatal rejection ceiling, immutable trainer snapshots, deterministic cross-rank sampling, and credit only after durable append keep targets and optimizer progress auditable under load. These are correctness and throughput properties, not direct chess-strength ablations; Chapter 5 covers the implementation.

4.2.9 What the retained curriculum establishes

The final data recipe combines staged replay growth, reuse four, policy-surprise sampling with a uniform component, unit row weights, shallow random openings, branch-reserved restart states, calibrated resignation, direct cut-position bootstraps, explicit target eligibility, coordinated publication, and partial actor/trainer overlap. Most of these choices coexist in one successful system and do not have one-variable online Elo estimates.

The strongest direct comparison favors a searched value over the tested material heuristic at a cut position. The strongest incident evidence is the late-game poisoning chain. The clearest systems lesson is that more presentations are not necessarily more information: replay reuse, target freshness, state diversity, and schedule pace move together. The retained design spends compute on a broad, inspectable stream of searched positions and preserves the meaning of every admitted target. It should be understood as an integrated curriculum, not a collection of additive strength bonuses.

4.3 Choosing what the network predicts

A network that fits self-play targets well but evaluates too slowly leaves the search without enough games to learn from. The retained design therefore joins two requirements: useful chess representations and inexpensive leaf evaluation. Policy geometry, global context, auxiliary targets, quantization, and model size each affected that balance. The evidence below distinguishes controlled component tests from the strength of the assembled system.

The model receives 52 side-to-move-canonical board planes: pieces, castling rights, en passant, checks, repetition, the eight most recent moves, material counts, and the fifty-move counter. File reflection is the only augmentation; it also mirrors action targets and exchanges kingside and queenside castling planes. Rule-sensitive planes prevent positions with different legal or draw states from becoming indistinguishable. Their individual strength effects were not ablated. Some component comparisons used an older 29-plane input, so their ab-

solute scores are not input-matched to the final model.

4.3.1 Three policy representations

The retained chess interface defines 1,880 canonical actions. Three implemented policy families represented move logits differently: a dense action projection, spatial move planes, and scores for origin–destination square pairs.

The **dense head** flattened a small spatial projection into 1,880 logits. Its tested variants changed projection width, spatial reduction, and the rank of the final map. A rank-96 variant reduced head size from roughly 484,000 to 207,000 parameters and was recorded as matching its immediate baseline. The result bundle is unavailable, so those variants cannot be ranked quantitatively here. Dense heads trained successful models; they were superseded, not shown intrinsically weak.

The **move-plane head** represented 56 sliding directions, eight knight moves, and 12 promotions in 76 spatial planes. One implementation gathered 1,880 canonical logits from the plane tensor; another exposed all 4,864 plane-square cells and normalized only legal moves. Native mapping checks covered 83,651 moves without a discrepancy, and inference controls did not attribute the slowdown to the larger action interface. In a short supervised comparison, the repaired plane head reached 2.1828 held-out policy cross-entropy after about 2,500 steps versus 2.0824 for the dense control, while its curve was still improving faster. Neither convergence nor a clean online strength comparison was established. An online plane-head trial was also retired after slower learning and weaker play, but its result artifact and exact representation have not been recovered. The plane approach is technically viable; its relative playing strength remains unquantified.

The retained **from-to head** preserves square structure without predicting a mostly empty plane tensor. It projects the 64 trunk squares into query and key vectors, scores all origin-destination pairs, and gathers the canonical actions through a fixed table. A separate projection adds queen, rook, and bishop offsets for promotions; en passant and the canonical castling encoding remain ordinary square pairs. On the controlled convolutional trunk, this head used 51,072 parameters rather than 483,680 for the dense alternative.

Holding that trunk fixed, the from-to head improved the held-out policy gap by 0.0298 nats (paired 95% interval 0.0285–0.0311). Spending the saved parameters on a wider trunk added only 0.0018 nats in the three measured cells; the missing fourth cell prevents a full factorial conclusion. The head cost approximately 1.9% of batch-512 forward throughput and 9% at batch 64. This is the strongest head-specific evidence in the project, though it is a short, single-seed teacher-data comparison, not an isolated self-play Elo ablation. We therefore report the policy fit improvement without converting it into a playing-strength estimate.

4.3.2 Convolution, attention, and global context

The retained convolutional residual tower is shared by policy, value, and training-only heads. The project did not test separate policy and value trunks. On the smallest networks, an oversized dense policy head and a second policy-shaped auxiliary consumed a large fraction of the parameters; that was a head-capacity problem, not a trunk-sharing experiment.

Pure attention trunks used 64 square tokens, learned row

and column embeddings, pre-normalized self-attention, and GELU feed-forward blocks. No-bias, relative-offset, and input-dependent Smolgen-style attention biases were implemented; only no-bias and Smolgen received a preserved efficacy comparison. A CNN-attention hybrid was proposed but not built.

Once bootstrap policy shape, head, runtime, and precision were controlled, the convolutional trunk beat bare attention by 0.0060 nats on held-out teacher data. Smolgen gave the best attention cell, but the larger effect was head choice: replacing the dense head on the attention trunk with from-to improved the held-out gap by 0.1573 nats. Attention also consumed more memory and served more slowly at the relevant batches. Convolution won this workload; the comparison does not establish a general limit of attention in chess.

Local convolution receives global context every second residual block: board-wide means and maxima from one quarter of the channels are projected back as biases on local features. Squeeze-excitation was another implemented context mechanism. Global pooling appeared to learn faster, with no clear final-strength difference, but the comparison artifact is unavailable. Its retention draws on that experience and the KataGo precedent, not an isolated chess Elo result. It also creates floating-point islands in the quantized graph: useful context and inference efficiency did not align perfectly.

4.3.3 Value and training-only heads

The outcome head predicts win, draw, and loss rather than a single scalar. Search converts this distribution to an expected value when necessary, while training and diagnostics retain the draw probability. A compact two-channel spatial reduction and 48-unit hidden layer was retained. A matched 32-channel probe added 97,020 parameters and reduced measured training throughput by 1.31%, while improving total loss by only 0.00309 in one short seed and slightly worsening WDL loss. This justified keeping the smaller head, not a claim that its capacity is universally optimal. Likewise, the earlier scalar-to-WDL transition has no preserved isolated strength comparison.

Training also predicts a later player’s policy at a configured ply offset and normalized remaining game length, with loss weights 0.15 and 0.1. The later-policy target uses the future state’s own side-to-move action space, legality mask, and symmetry transformation. Both heads are removed from the serving artifact. Their target wiring, masking, gradients, and checkpoint behavior were tested, but neither objective has a long matched self-play ablation. Other auxiliaries were set aside while debugging potential interference, not shown harmful. Their inclusion in the final recipe should not be read as an isolated Elo gain.

4.3.4 Bootstrap and optimization controls

Initialization shapes the first search priors and, through them, the first self-play targets. In an initial architecture comparison, the attention policy put only about 0.11 probability mass on its top three moves while the convolutional control was effectively one-hot. Neither extreme represented chess knowledge, but each induced different data. The retained bootstrap uses architecture-appropriate initialization, deterministic construction, a small final policy projection, and calibration on

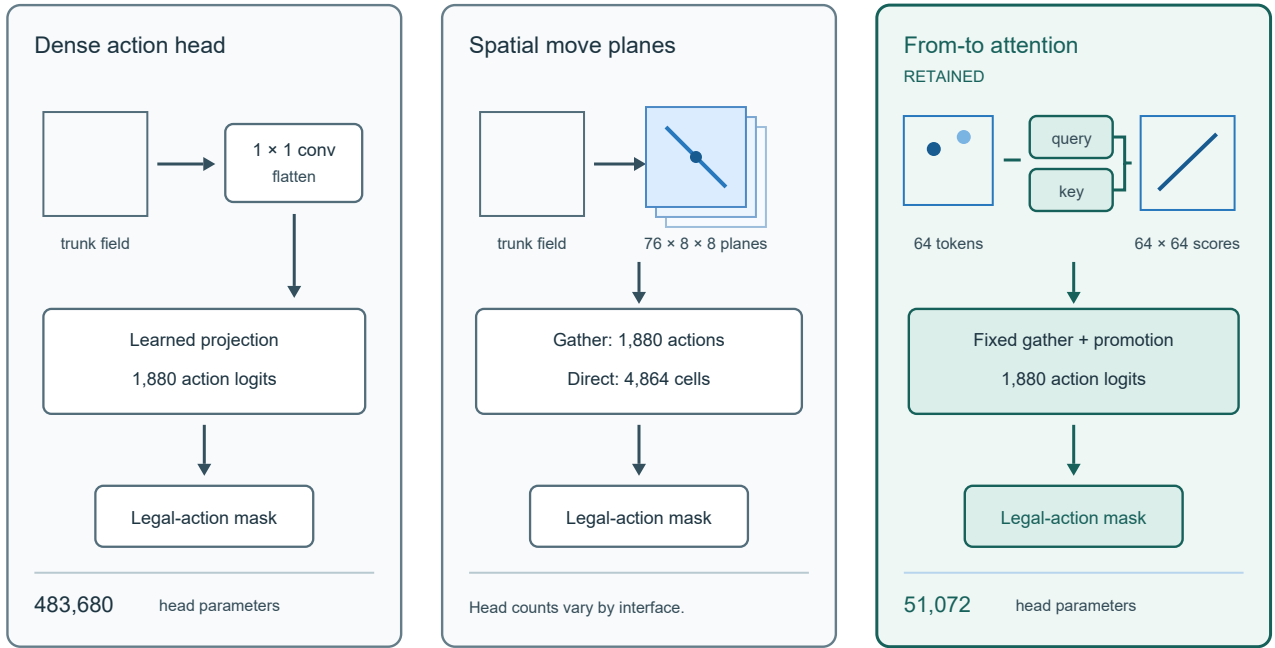


Figure 4: Policy representation changes the learned output geometry. Dense heads learn an action-sized projection, move-plane heads preserve spatial move types, and the retained from-to head scores square pairs before a fixed gather. The parameter counts are from the same controlled convolutional-trunk comparison; no count is shown for the plane family because its implementations and external interfaces differed. The diagram is architectural, not a playing-strength comparison.

516 encoded positions toward a common policy concentration. Those positions set numerical scale, not supervised chess targets. A seed-propagation defect also made nominally matched arms start from different tensors; corrected comparisons treat seed and policy shape as controls, not strength improvements.

The retained quantization-aware optimizer is Nesterov SGD. Frozen-replay screens showed stable fitting and informed warm-up, learning-rate, and folding choices; they did not measure online Elo. The selected run ended at 408,500 optimizer steps, below its configured one-million-step folding trigger: the trainable model remained pre-fold, and folding was performed on a deployment copy. Delayed folding and higher post-fold rates helped short-horizon replay fitting in separate screens; they were not a phase of the reported training run. AdamW had trained successful earlier models and was superseded, not disproven. Frequent gradient clipping did not prevent learning in those screens, but its threshold was not shown optimal.

4.3.5 Quantization as an architectural constraint

Post-training quantization of the ordinary residual tower was fast and behaviorally unusable. Activation ranges grew from about 0.78 near the input to roughly 40–43 late in the network; depending on calibration, full-trunk INT8 preserved only 13.3–25.9% policy top-one agreement. Weight-only quantization was faithful but slower than TensorRT FP16. Quantization therefore became a network-design problem rather than a final export switch.

A scaled pre-activation block bounded the learning graph, but its normalization, clipping, residual scaling, and requantization boundaries fragmented the compiled TensorRT graph. It expanded an 83-layer FP16 graph to 330 layers and an INT8 graph to 470 layers, with many reformats, yet still failed

fidelity. The retained scaled post-activation block instead keeps the conventional sequence of convolution, normalization, capped activation, convolution, normalization, scaled residual addition, and capped activation. Activations are capped at six, and residual branches are scaled by the inverse square root of depth. The scale can be folded into the second convolution at export, which preserves more efficient compiler tactics.

This block learned normally under quantization-aware training. After continuation in the folded deployment topology, a production-sized smoke test reached roughly 135,000 INT8 positions per second, compared with 60,000 for TorchScript BF16 and 99,000 for TensorRT FP16. These are model-core rates, not end-to-end self-play rates. Folding only after training damaged agreement, and global context and the heads remain outside the INT8 trunk. The result is evidence that architecture and deployment had to be co-designed; it is not evidence that the scaled block plays better than an ordinary residual block in floating point. The fidelity and throughput measurements above distinguish the tested alternatives.

4.3.6 Progressive model sizing

KataGo provided the precedent [2]: start with a small, fast network, train the next size on the same data, and switch when it catches up. Early in self-play, extra model capacity may contribute less than the additional searched games a small network can produce. The measured small-model throughput supports that premise here, and the small-to-medium handoff worked repeatedly. There is no equal-cost fixed-size control, so its exact Elo-per-currency contribution remains unknown.

Candidate start follows a stage-specific searched-Elo plateau; promotion instead requires two passing paired matches against the active model. The former loss-based gate



The medium model is the reported checkpoint; the larger stage is a capacity hypothesis, not a demonstrated improvement.

Figure 5: The small model buys early self-play throughput, and a medium candidate trains on the same replay before paired-match promotion. The larger candidate may avoid catch-up with function-preserving growth, but the limited continuation did not demonstrate a strength gain; the reported checkpoint remains medium-sized.

promoted a candidate that was about 270 Elo weaker because extra catch-up updates made its training loss incomparable. The failure study in Chapter 6 explains that correction; Chapter 7 states the retained thresholds and promotion gate.

The larger stage remains unresolved. An independently initialized candidate needed substantial catch-up. Explicit function-preserving growth avoided relearning the medium model’s function, but may also bias optimization toward its existing representation; that possible capacity cost was not measured. The limited grown-model continuation reached parity without a clear strength gain, so the reported model remains medium-sized. A matched-compute comparison of independent catch-up and growth would be needed to choose between them.

4.3.7 Distillation and compact models

Two compression studies asked related but distinct questions. Teacher-output imitation trained compact students on a legal-masked policy and WDL distribution from policy-only teacher play; no search generated those labels. Increasing the dataset from one million to six million positions mattered more than a small capacity sweep. The strongest 1.33-million-parameter student trailed its teacher by 176.1 Elo at 25 searches each and by 38.4 Elo when it received the measured shallow equal-compute allowance of 58 searches. At 250 searches each the gap widened to 257.6 Elo. Deeper search amplified the better prior rather than washing out approximation error. Fixed-ply sampling gave the dataset only one side-to-move parity, and the teacher had a known conversion weakness, so these absolute gaps do not transfer to the final model.

Replay-target compression instead trained on sparse MCTS visits, outcomes, root values, and replay metadata from a frozen ten-million-row window. The selected 474,069-parameter student was 13.20 times smaller than its teacher. It trailed by 291.3 Elo at 64 searches each and by 166.2 Elo when its measured saturated serving advantage allowed 186 searches against 64. It reached statistical parity only at an equal-multiply-accumulate allowance of 850 searches, which ignored tree work, launches, and imperfect batching and was therefore not an equal-time result. The compact artifact was useful, but it did not replace direct training. These were separate compression protocols, not direct comparisons with the

final teacher.

A terminal compression check trained a 470,295-parameter student on a separate frozen 20-million-row replay snapshot. At 10,000 searches it reached 2,683 conditional benchmark Elo after roughly 7.5 epochs and 2,697 after roughly 23 epochs. The 14-Elo central difference lay well inside the match intervals, while held-out policy loss had become nearly flat. Tripling passes over this fixed buffer therefore produced no measurable playing gain. The longer student reached 2,873 conditional Elo at 100,000 searches, but the point is unbracketed and is not a calibrated deep-search headline.

Together, the studies show that a small student can preserve substantial behavior and make a practical published artifact, but not that model size can be exchanged mechanically for more search. Realized search multipliers were far below parameter or arithmetic ratios, and the teacher advantage often grew with search depth. Distillation is a compression result, not a stage of the primary self-play algorithm.

4.3.8 Decision

The retained network combines the rule-complete 52-plane input, a shared convolutional trunk, periodic global context, the from-to policy head, a compact WDL head, and training-only next-policy and remaining-length objectives. Scaled post-activation blocks make the trunk compatible with quantization-aware deployment. Progressive sizing successfully exploited a small model before handing off to the medium model, while the value of the larger stage remains unresolved. This is one integrated design supported by component tests of different strength; the report does not assign isolated Elo gains where only proxy, throughput, or assembled-system evidence exists.

5 From inference speed to learning speed

Search-based learning needs enough completed games to supply the trainer. Under a fixed hardware budget, the useful systems objective is therefore admitted self-play positions and strength per wall-clock hour, not model evaluations per second in isolation. The gap between those quantities determined where the engineering effort mattered.

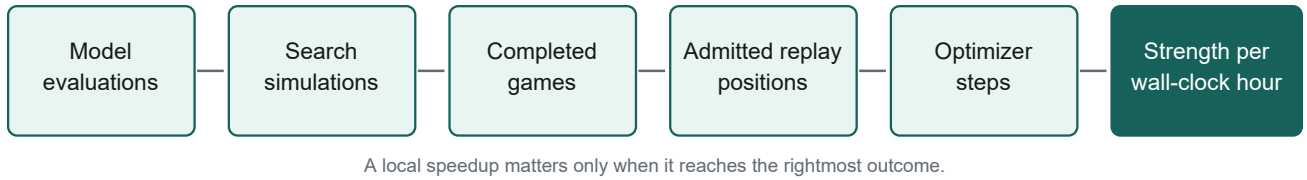


Figure 6: The denominator changes at each boundary. Model evaluations and search simulations describe local service capacity; completed games and admitted rows determine training supply; only the final stage measures the playing-strength gain per unit wall-clock time.

5.1 Native ownership of the search loop

Python coordinates configuration, processes, replay, training, evaluation, and publication. C++ owns game rules, encoding, search trees, selection and backup, and asynchronous inference requests and results. This division keeps Python message serialization and per-position dispatch off the dominant self-play path. The earlier Python search loop is a performance baseline, not an alternative production engine.

The native boundary also enables batching across many simultaneous games. Each self-play process advances 512 game roots, and requests from those roots share a preallocated asynchronous inference pipeline. Trees survive played moves so previous work can be retained rather than reconstructed. The final topology uses four actor processes per GPU, one inference worker per process, batches of at most 320 positions, and two outstanding batches. Those settings are a joint CPU, memory, batching, and latency choice; they should not be interpreted independently.

5.2 Batching and host-side submission

Large neural batches help only while the host assembles, submits, and consumes them fast enough. Board encoding, persistent staging buffers, host-to-device copies, result processing, CUDA events and graph replay, process count, and inference-thread count were optimized as one submission path.

In a controlled 32-process self-play workload, the optimized path increased search throughput from 512,679 to 617,782 searches/s, a 20.5% gain, while reducing aggregate actor CPU consumption from 52.6 to 19.8 cores. Average inference batch size rose from 141 to 222 and the number of model calls fell by 24%. On the same node restricted to 24 CPU cores, throughput increased from 215,265 to 496,036 searches/s because reducing submission overhead allowed the previously starved GPUs to remain busy. These figures are specific to the tested network and host, but they show why CPU efficiency can be decisive even in a nominally GPU-bound workload.

Batch fill alone did not guarantee device saturation. With the TensorRT actor, reducing four processes per GPU to two lowered simulations/s by 39% even though both filled individual batches: concurrency overlapped CPU and GPU work around inference. Additional inference threads had the opposite effect, duplicating CUDA contexts and splitting requests into smaller batches. One inference thread per actor was the measured operating point.

Search parallelism is a separate batching lever. Multiple leaves from one root can fill a batch after other games finish, but virtual reservations make the search less serial and can reduce search quality. Chapter 4 therefore treats parallel search

as an algorithmic quality-throughput trade, not a free systems optimization.

5.3 Inference runtimes and precision

TorchScript supplied the trimmed policy/WDL bootstrap and fallback artifact, making it the relevant comparator to a new serving runtime. `torch.compile` sped up eager batch-64 inference by roughly 27–33%, but fused TorchScript was faster than the compiled eager path. In the tested eight-GPU training workload, compilation reduced throughput by about 18% relative to eager execution; `bfloat16` autocast improved it by 9.2%. Compilation was not retained for production inference or training on this workload.

TensorRT quantization and engine refitting [8] provided the stronger serving path. A native FP16 engine reached 75,889 positions/s against 40,716 for the TorchScript BF16 control in one matched backend benchmark, a 1.86x ratio. Quantization-aware INT8 inference added a further 1.31x over the TensorRT FP16 engine for the tested quantization-oriented network. Matched production-topology comparisons measured smaller but still material INT8 gains that depended strongly on model shape: 14.4% for the smaller network and 39.1% for the medium network.

Those gains depended on model design: post-training INT8 was fast but changed policy and value outputs too much. The retained scaled post-activation blocks are trained with fake quantization. Backbone convolutions run in INT8; the start block, heads, and linear layers remain at higher precision. Published checkpoints are recalibrated, exported as explicit Q/DQ ONNX graphs, and refitted into shape-specific TensorRT templates. Legal-move policy and WDL fidelity are checked on encoded positions, because a built engine is not necessarily a faithful one. Section 4.3 describes the block design; Chapter 6 describes a refit failure that made semantic checks necessary.

Model shape remained a systems variable even at similar parameter counts. Width and depth changed kernel efficiency, TensorRT tactics, and memory behavior discontinuously. Channels-last layout and cuDNN autotuning helped relevant CNN shapes, but no analytic parameter-count rule predicted the fastest network. Progressive model sizes were therefore benchmarked at their actual serving batch and precision rather than selected from FLOPs alone.

5.4 Replay materialization and training supply

Search work becomes useful training compute only after games finish and their observations pass replay admission. Completed games are atomically published, converted into typed columnar rows, and appended to a fixed-capacity memory-mapped replay. Materialization is partitioned before

expensive conversion, bounds inbox scans and staging, quarantines malformed individual games, and fails loudly if systemic rejection prevents the learner from receiving data. This makes replay ingestion both a throughput stage and a scientific boundary.

The trainer reads vectorized batches from the memory-mapped store with pinned-memory prefetch and one persistent DDP rank per GPU. A loader benchmark on 2.5 million rows improved from 3,943 to 32,579 samples/s after compacting 5,000 small producer shards into 25 containers—8.26x faster and 44% above the measured trainer demand. In a later live interval, materialization appended 8,790 positions/s while accepted positions arrived at 1,365/s, leaving more than sixfold headroom. Materialization was not the bottleneck in that workload.

A credit ledger links optimizer work to newly admitted positions. Each training quantum becomes eligible only after enough materialized data has arrived for the configured replay-reuse ratio. This prevents rejected games, ingestion stalls, or an accidental change in data reuse from appearing as normal optimizer progress. Persistent DDP ranks then avoid repeated startup and model construction costs. On the measured eight-GPU topology, a global batch of 2,048, bfloat16 autocast, and concurrent self-play produced 6,252 training samples/s; larger batches improved hardware throughput further but would also change update count and optimization semantics, so they were not adopted merely because the GPU benchmark was faster.

5.5 Overlapping self-play and training

Self-play, materialization, training, publication, and evaluation overlap without making the learner fully asynchronous. Optimizer work remains divided into explicit quanta, and actors switch models only at publication boundaries. A selected subset keeps producing games during training while the rest pause. This uses idle capacity without losing the identity of the model and data behind each quantum.

The overlap fraction is a measured compromise. With no actors running, the trainer processed 25,275 samples/s. Keeping 8, 16, or all 32 actors active reduced trainer throughput to 21,492, 17,139, and 9,210 samples/s, while simultaneously producing approximately 506,000, 606,000, and 742,000 searches/s. When both training time and the remaining self-play wait were included, the estimated complete-quantum times were 117.1, 112.9, and 111.2 seconds. The broad result was that moderate overlap improved total cadence, while running every actor nearly doubled the trainer quantum for only a small additional end-to-end gain. The retained half-active policy sits near that measured tradeoff rather than maximizing either trainer or actor throughput alone.

The useful overlap fraction depends on search cost. As visit budgets and model size rise, self-play becomes more expensive relative to training; actor settings measured at the beginning cannot simply be extrapolated to later stages.

5.6 Where throughput becomes useful

Faster simulations did not translate mechanically into training throughput. In a production-shaped 400-visit benchmark, a TensorRT INT8 actor executed 1.86 times as many simulations as a floating TorchScript control with the same architecture and configuration. Their weights differed, so this is a backend-and-checkpoint comparison, not an isolated preci-

sion effect. A separate live 400-visit stage admitted 23.3% more replay positions per second than a 600-visit stage. Visit budget, checkpoint, and actor scheduling differed between those stages; the 23.3% is not a measured downstream effect of the 1.86x benchmark. Faster actors may finish shorter games, checkpoint publication interrupts work, and replay credit appears only after complete-game materialization. In the live stage, admitted-position rate predicted optimizer cadence exactly.

Inference and search benchmarks diagnose a mechanism. Completed games, accepted replay, optimizer cadence, and learning curves show whether that mechanism saved useful training time.

Finally, absolute throughput is not portable across machines. CPU quota, GPU power limits, PCIe and NUMA layout, driver/runtime versions, model shape, batch fill, and concurrent training all affect the operating point. Node comparisons informed hardware selection, but quantitative claims in this chapter remain attached to the documented hardware and workload.

6 Three failures that changed the method

Three plausible shortcuts failed at different boundaries: which positions entered replay, whether an inference engine represented its source model, and whether a larger candidate was ready to replace the active model. Their causal sequence matters because the apparent success signal in each case measured the wrong thing.

6.1 Late-game target poisoning

The self-play design once reduced search near the end of long games. The intent was to keep noisy positions from exceptionally long endgames from dominating replay. If those positions would not become training targets, spending full-search compute on them seemed wasteful, so the tail used cheap searches instead. Only full-search positions were admitted as primary training rows, however, so the cheap-search tail selectively removed the conversion phase from the training set. The resulting policy was weakest exactly where it needed to turn an advantage into a terminal result. Games wandered to the ply cap more often, and the cutoff then assigned one terminal value to every earlier training row in the trajectory.

This created a closed feedback loop. Missing searched endgames produced weak late-game play; weak play produced drawn-out, unconverted games; the cutoff supplied an unreliable target; and that target reinforced the same policy and value errors. During the most affected early interval, roughly one ply in seven was structurally ineligible for training, 27–32% of games reached the cap, and about 36–38% of admitted rows inherited the value assigned to a cut game. The corruption was therefore not confined to the last position: the final target propagated back through the entire recorded trajectory. By the time the active replay buffer looked healthy, the damaging rows had already been evicted, while their effect on the weights and subsequent self-play distribution could remain.

The repair happened in two stages. First, the cutoff stopped relying on the material heuristic and obtained a value from one full search at the final cut position. A controlled continuation study supported that choice: on 2,282 early cut positions, the searched root value reduced Brier error from 0.491

to 0.444 and cross-entropy from 0.851 to 0.756 relative to the material target. The same ordering held on a later set of 1,144 positions. Later, the forced cheap-search tail was removed altogether, returning properly searched endgame positions to replay. The second change closed the data hole that a better terminal value alone could not repair. Because other late-game changes were bundled around the same period, neither step receives an isolated Elo credit. The evidence establishes the failure mechanism and target-quality improvement, not a one-variable strength estimate.

Search eligibility, replay admission, and terminal-value provenance must be designed together. Weakening or discarding one part of a trajectory can make the model least competent there; subsequent self-play then reinforces the same blind spot.

6.2 A mechanically valid but semantically invalid TensorRT refit

The deployment pipeline builds a structural TensorRT template and refits it with each new quantized checkpoint. One template was built from a checkpoint whose clipped activation quantizers had many numerically equal scales. At high optimization levels, TensorRT optimized the engine around those equalities. Training later separated the scales, but the refit API still accepted every replacement weight, reported no missing weights, and returned success. The resulting engine was structurally valid and semantically wrong.

The defect was visible only when the published engine was treated as a chess model rather than as a successfully serialized object. On 516 real positions, the faulty template produced legal-policy KL near 1.05 and repeated refits of the same inputs changed individual logits by 10–15. Building at a lower optimization level, or first making the template’s scale constants distinct, reduced legal-policy KL to about 0.0012 and made repeated refits deterministic. The failure required three conditions: equal scales in the build source, optimization level four or higher, and a later refit that made those scales unequal. Tests with identity batch normalization and zero biases rejected constant folding as the cause. These probes isolated the refit-template defect.

Template construction now separates equal quantization scales before optimization and uses a safer default optimization level. More importantly, deployment publication no longer treats deserialization, complete weight accounting, or refit success as evidence of model equivalence. It evaluates the engine on real encoded positions and records legal-move top-one agreement, legal-policy KL, and WDL error against the source artifact. Those checks cover the semantics that search actually consumes: which legal move the policy prefers, how its probability mass changes, and whether the value distribution remains faithful.

A compiler or refitter can meet its mechanical contract while violating the model’s behavioral contract. Published artifacts therefore require representative behavioral probes alongside their graph, template, runtime, and refit manifest.

6.3 Promotion from incomparable training losses

The progressive controller originally promoted a larger candidate when its smoothed training loss caught up with the active model. Loss looked like a cheap proxy for readiness: it was already measured during training, whereas a full-search

candidate match consumed additional evaluation compute and time. That comparison ceased to be meaningful once the candidate received extra optimizer quanta over the same replay distribution. More presentations gave it a systematic advantage on replay loss without proving equal playing strength. The loss gate promoted a candidate whose deployed artifact passed inference-fidelity checks but was still about 270 Elo weaker in play.

Promotion now measures the property that publication requires. The candidate plays paired head-to-head matches against the active deployment artifact and must score at least 0.48 in two consecutive completed evaluations. A failing score resets the sequence; a failed or cancelled match contributes no evidence. Candidate-start timing, extra catch-up training, and promotion are separate controls. Function-preserving growth is likewise a separate attempt to remove the larger model’s initial relearning deficit, not a substitute for the playing-strength gate. Chapter 7 states the retained candidate-start and promotion procedure.

Training loss remains valuable for optimization diagnostics. It is not a promotion criterion when the compared models have seen different numbers of presentations or when the artifact intended for deployment can be tested directly.

7 Integrated chess recipe

7.1 Living recipe and frozen result

The readable entry point for reproducing or extending the system is the fully expanded `chess-final-config.yaml` [10]. It owns the current network ladder, trainer, replay, self-play, deployment, and online-evaluation settings without an inheritance chain. It is a living recipe: future project work may revise it.

The reported experiment is immutable. Chapter 8 and Appendices B and D identify the selected checkpoint, evaluation rows, and provenance boundary. These frozen facts—not a future state of the editable YAML—are the authority for published numbers.

The reported node had eight NVIDIA GeForce RTX 4070 SUPER GPUs and 80 logical CPUs. Hardware-sensitive throughput comparisons in this paper refer to their stated workloads on this class of node, not a portable rate guarantee.

7.2 Model, objective, and optimization

The configured ladder contains 12×128, 14×160, and 19×176 residual CNNs. Every stage uses capped scaled post-activation branches, global-pooling context in every second block, a key-size-128 chess from-to policy head, and a two-channel WDL value head with a 48-unit hidden layer. Training-only heads predict the next searched policy at weight 0.15 and remaining game length at weight 0.1; deployment removes both.

The primary policy and value losses each have weight 1.0. Terminal outcome targets are discounted by 0.998 per ply. A search-root-value blend rises from zero to 0.1 over its configured schedule, while search backup uses a separate 0.99 per-ply discount. Cut games use the searched root value and censor unknown remaining-length labels.

Each training quantum contains 500 optimizer steps. Eight ranks each process 256 positions for a global batch of 2,048.

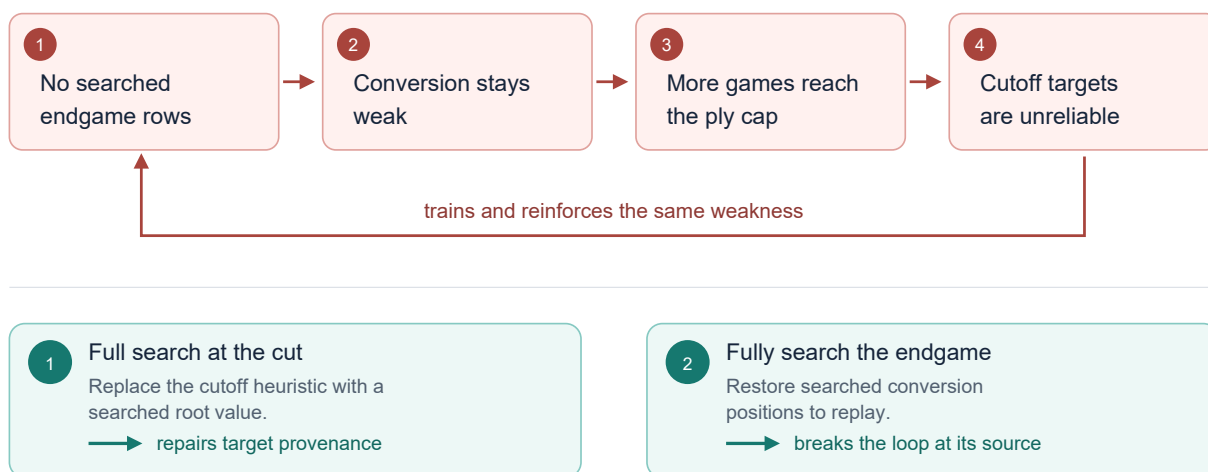


Figure 7: The cutoff repair fixes target provenance; restoring fully searched endgames breaks the feedback loop at its source.

The optimizer is Nesterov SGD with momentum 0.9 and weight decay 0.0001; gradients are clipped to norm 1.0. The learning rate warms from zero to 0.1 over the first 1,000 optimizer steps and then follows the configured linear schedule to 0.01. Training uses bfloat16 and persistent trainer processes; `torch.compile` is disabled.

INT8 quantization-aware training is active from the beginning. Calibration uses 516 real evaluation positions and is refreshed at every publication boundary. The trainable model remains pre-fold; deployment folding, recalibration, ONNX export, and TensorRT refitting operate on a copy. This keeps serving conversion from changing the optimizer’s parameterization.

7.3 Self-play, replay, and curriculum

Self-play runs 32 actor processes, four assigned to each GPU, with 512 interleaved games per process. During a training quantum, half of the actor processes on every GPU pause while the others continue producing games. Native inference uses batch 320 with two outstanding batches per worker.

The fixed search budget grows from 300 to 800 visits per move. Search uses exploration constant 1.5, reduced-parent FPU with reduction 0.2, forced playout coefficient 1.5, Dirichlet epsilon 0.25, and alpha 0.3. Fixed visits were retained after adaptive allocation and learned stopping failed to improve wall-clock strength.

The configured draw assigns half of games up to eight random legal plies and half to recent restart states filtered by value, remaining length, age, and branchable visit mass; missing restart candidates fall back to random openings, so this is a configured rather than guaranteed realized 50/50 mix. Restart selection retains a 30% uniform component. Game caps grow from 150 to 250 plies, and greedy move selection begins later as training matures. Calibrated resignation begins only after sufficient evidence, constrains the false-nonloss upper bound to 2.5%, and designates 20% of games at creation as no-resignation continuations for ongoing safety measurement.

Replay capacity grows through ten stages from 600,000 to 20 million live rows. Each admitted position funds four training presentations. Sampling reserves 30% uniform probabil-

ity and otherwise prioritizes bounded policy surprise. Eight materializers convert completed games into the fixed-layout memory-mapped store; credit is committed only against durable admitted data.

7.4 Progressive sizing

Candidate start and promotion are separate decisions. The primary 64-search ladder is smoothed and used to detect stage-specific plateaus. The current configuration [10] specifies the thresholds and window. An eligible successor receives an average of 1.5 optimizer quanta per active-model quantum on the captured replay snapshot and uses its own catch-up learning-rate clock.

Promotion is decided by candidate-versus-active matches, not training loss. The configured gate requires a score of at least 0.48 in two consecutive paired evaluations. This distinction matters because extra candidate presentations made the earlier loss comparison systematically favorable to the candidate and admitted a much weaker model.

The completed capacity study also grew the trained 14×160 network into 19×176 while preserving its function, then recovered deployment fidelity through QAT. The larger continuation reached parity but did not establish a stronger plateau. The reported checkpoint is therefore the 14×160 model. The limited continuation cannot distinguish insufficient training or post-growth optimization from a target, replay, or capacity limit.

7.5 Training-time and terminal evaluation

During training, evaluation ran every 20 minutes on 50 paired openings. The searched ladder used 64 searches per move against a three-rung adaptive bracket of fixed-node Stockfish 13 opponents; a policy-only ladder used the same opening and opponent framework. These measurements controlled progress and candidate timing but are distinct from the terminal result matrix.

The terminal matrix evaluated the selected 14×160 checkpoint over 100 games per row from 50 paired openings, with each opening played from both colours. Stockfish 13 used one thread, 1,024 MiB hash, and fixed node budgets from the published anchor curve. The reported rung for each model

budget is the one whose observed score is closest to 0.5; both rungs and confidence intervals appear in Appendix B. Searched play used the frozen INT8 TensorRT artifact. Policy-only play used the matching float TorchScript export because it bypasses the native search service.

The measured curve spans policy only and 100, 1,000, 10,000, and 100,000 searches per move. Parallelism is one at 100 and 1,000 searches, four at 10,000, and sixteen at 100,000. It is therefore an attainable operating curve, not a single-variable search-scaling experiment. Absolute values are protocol-specific benchmark Elo and must not be presented as FIDE ratings or unrestricted engine ratings.

7.6 Interpretation

The terminal matches evaluate this assembled recipe; they do not divide its strength among replay growth, restart states, auxiliary targets, resignation, progressive sizing, and the other retained choices. Chapters 4–6 state which components have separate strength, fidelity, throughput, or correctness evidence. Chapter 8 reports the selected model’s outcome and the narrowly scoped training cost.

8 Final training and evaluation results

The selected 6.32-million-parameter chess model reached 3,251 benchmark Elo at 100,000 searches per move after 2.5 days of final-run training. Its terminal strength is measured by paired games against a fixed-node Stockfish 13 ladder, not by the training ladder alone. This chapter reports that operating curve, the training volume behind it, the matched-estimator comparison with the previous baseline, and the limits of the student and parallel-search results. Appendix B contains every terminal match row.

8.1 What was selected

The strongest fully retained checkpoint is a **6.32-million-parameter, 14-block, 160-channel convolutional model** with scaled post-activation residual blocks, global context conditioning, and a chess from-to attention policy head. It was trained with quantization-aware training. Searched results use its INT8 TensorRT artifact; policy-only results use the corresponding float TorchScript export because no native search service is involved. The inference path is therefore stated with each result rather than silently treating unlike back-ends as identical.

The ladder’s strongest region spans several nearby checkpoints rather than a single isolated spike. Checkpoint 1026 was the last checkpoint in that region for which the full model, optimizer, QAT, ONNX, and TensorRT set had been retained. A subsequently grown 19-block, 176-channel model recovered its parent’s strength but remained flat and did not justify replacing the selected checkpoint.

8.2 Training outcome

The report-scoped final lineage contains **180 observations through exactly 2.5 effective days**. It rose from 798 to **2,372.2** at the cutoff and reached a peak of **2,407.6** inside that interval. Later points belong to capacity and training experiments that did not improve the accepted result and are excluded from the comparison rather than presented as continued final-model training.

The previous four-day baseline is cut at exactly **3.0 days**,

retaining 143 clean observations and ending at 2,265.4 Elo. Later points from its noisy terminal evaluation interval are excluded. The two plotted endpoints differ by 106.8 Elo, and the final endpoint is 348.2 Elo above the early baseline’s terminal observation. These are descriptions of the trimmed curves, not valid cross-campaign strength estimates, because the previous baseline used a single-rung fit while the final campaign’s generic ladder series used a three-rung bracketed fit.

A retrospective plateau comparison on the same three-rung estimator places the previous baseline at 2,283.9 Elo and the final recipe at 2,358.0 Elo: **+74.1 Elo**, with an approximately **±15 Elo transfer/sensitivity allowance**. The allowance reflects uncertainty in transferring the estimator correction; it is not a game-level bootstrap confidence interval. The report therefore uses **about +74 Elo under a matched estimator** as the cross-campaign headline and does not compare raw peaks. Repeating the arithmetic with only the report figure’s clean 3.0- and 2.5-day windows gives **+75.8 Elo**; the conclusion is not driven by the later points omitted from the plot. The plateau arithmetic and its transfer assumption are given in Appendix B.

The plotted lineage excludes reverted INT8 and capacity-promotion branches. The latter prompted the move from training-loss promotion to paired deployment-artifact matches, described in Chapter 6. A later function-preserving larger-model continuation reached parity without a demonstrated improvement. Its short horizon does not separate optimization difficulty, target quality, replay composition, and additional capacity.

The selected checkpoint records **408,500 completed optimizer steps**. With the configured global batch of 2,048 this corresponds to **836,608,000 training presentations**. The frozen coordinator events cover all 817 contiguous training quanta on that path. Summing their per-quantum ingested-game counters gives **3,249,647 completed games**; the credit ledger ends at approximately **209.15 million net materialized positions**, or about **4.00 training presentations per net materialized position**. Replay held **16 million live rows** at the selected checkpoint. These are selected-lineage counters, not totals for every experiment or every game generated on the node. The cumulative position scalar is float32 in TensorBoard, so its final few integer digits are not meaningful. Appendix A shows the training diagnostics and clarifies the boundary of each count.

The network-evaluation total is estimated rather than counted. Using a conservative 100 searched plies per completed game and roughly 600 simulations per ply gives about 195 billion search simulations across 3.25 million games, reported as an estimate in the abstract. Most require a neural-network evaluation; terminal leaves and reuse make this an approximate scale measure, not an exact forward-pass counter.

Across 480 small-model quanta, median measured trainer throughput was 16,977 samples/s; across 337 medium-model quanta it was 11,194. Model shape, schedule, and concurrent workload all differed, so this is not an isolated model-size effect.

The narrow cost attached to the selected checkpoint is **\$43.20**, calculated as 60 accepted-lineage hours at \$0.72/h. It excludes the reverted work, later growth experiment, distillation, evaluation, and idle rental time. Until total spend is

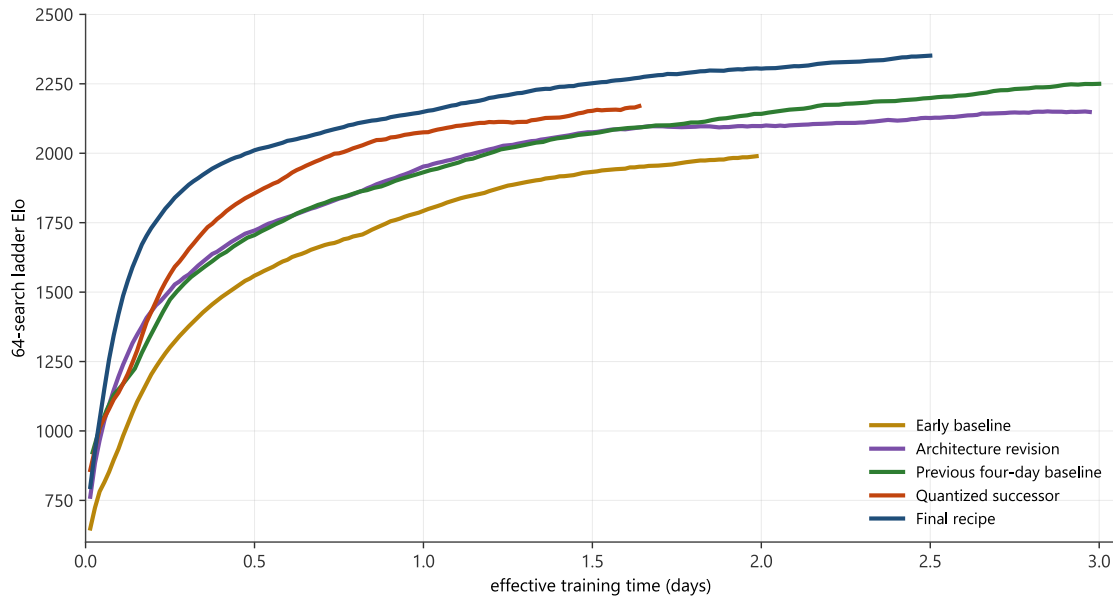


Figure 8: Each curve uses the bias-corrected 0.95 exponential moving average used by the project’s earlier training-dynamics plots. The final recipe is cut at 2.5 days and the previous four-day baseline at 3.0 days. The figure is descriptive; the +74.1 Elo claim comes from the matched-estimator plateau audit rather than from subtracting its displayed endpoints.

reconciled, it must not be described as the project’s total compute cost.

8.3 Strength across four decades of search

The terminal protocol used 100 games per row from 50 colour-swapped opening pairs against single-threaded Stockfish 13 at fixed node limits. For each search budget, the reported rating is the opponent rung whose score lies closest to 0.500. Anchor ratings come from Marco Meloni’s fixed-node Stockfish 13 benchmark [9], which connects Stockfish through Fruit 2.2.1 to the historical SSDF scale. The resulting values are benchmark Elo, not FIDE ratings or estimates on a current unrestricted-engine list.

The observed policy-only-to-deepest operating-point difference is 1,593 benchmark Elo, with smaller increments at each later search budget. This is not an isolated search effect: the policy-only and searched artifacts differ, and parallelism changes across searched conditions. The bootstrap intervals quantify match sampling conditional on the fixed, graph-read Stockfish anchor ratings; they do not include anchor-calibration uncertainty. The 100,000-search estimate is unusually well anchored: an independent 100,000-node opponent gives 3,247 Elo, only four points below the 200,000-node result. That local agreement supports the top headline, but does not validate extrapolation beyond the measured anchors.

Appendix B reports the complete ten-row W/D/L matrix and match-bootstrap intervals.

8.4 Two protocol effects that matter

Opponent rungs disagree at lower budgets At 100, 1,000, 10,000, and 100,000 searches, the harder opponent rung reads 128, 102, 46, and 4 Elo higher than the easier rung. Ideal transitive Elo would give the same estimate from both. Draw behavior, anchor calibration error, matchup effects, and sampling noise are possible contributors, but the project did not isolate the cause. Choosing the score closest to 0.5 limits extrapolation. The disagreement matters at low budgets but is

immaterial to the deepest result.

Parallelism buys time by spending strength The headline curve is an operating curve, not a pure search-budget ablation: it uses one parallel search at 100 and 1,000 searches, four at 10,000, and sixteen at 100,000. A controlled 1,000-search sweep measured 2,823 Elo with one parallel search, 2,804 with four, and 2,778 with sixteen. Four-way parallelism therefore cost 19 Elo; sixteen-way cost 45 Elo.

The two recorded timing totals give **18.1–18.5 minutes** with one parallel search, **3.4–3.8** with four, and **1.3–1.8** with sixteen. These are descriptive ranges across the operational recap and result manifests, not confidence intervals. The corresponding paired speedup ratios span approximately **4.8–5.3x** for four-way and **10.1–13.9x** for sixteen-way parallelism. At only 100 searches, sixteen-way parallelism cost 235 Elo, so parallelism cannot be changed silently across a compute curve. The available points suggest that a fixed parallel count becomes less harmful as the total budget grows, but they are too sparse to define how much parallelism is effectively free at each budget. That frontier would require a dedicated budget-by-parallelism grid.

8.5 What the student establishes—and what it does not

Both students have **470,295 parameters**, 13.4 times fewer than the teacher, and trained on the same separate, frozen 20-million-row replay snapshot. The selected teacher checkpoint held 16 million live replay rows; the student snapshot is not that checkpoint’s training window. At 10,000 searches against the same 20,000-node anchor, extending training from 36,621 to 110,000 steps moved the estimate only from **2,683 [2,637, 2,731]** to **2,697 [2,640, 2,753]** benchmark Elo. The 14-Elo central change lies inside match uncertainty, while the longer student’s held-out policy loss had nearly flattened. At 100,000 searches it reached **2,873 [2,819, 2,935]**, but that point is unbracketed. The student used TorchScript and the teacher INT8

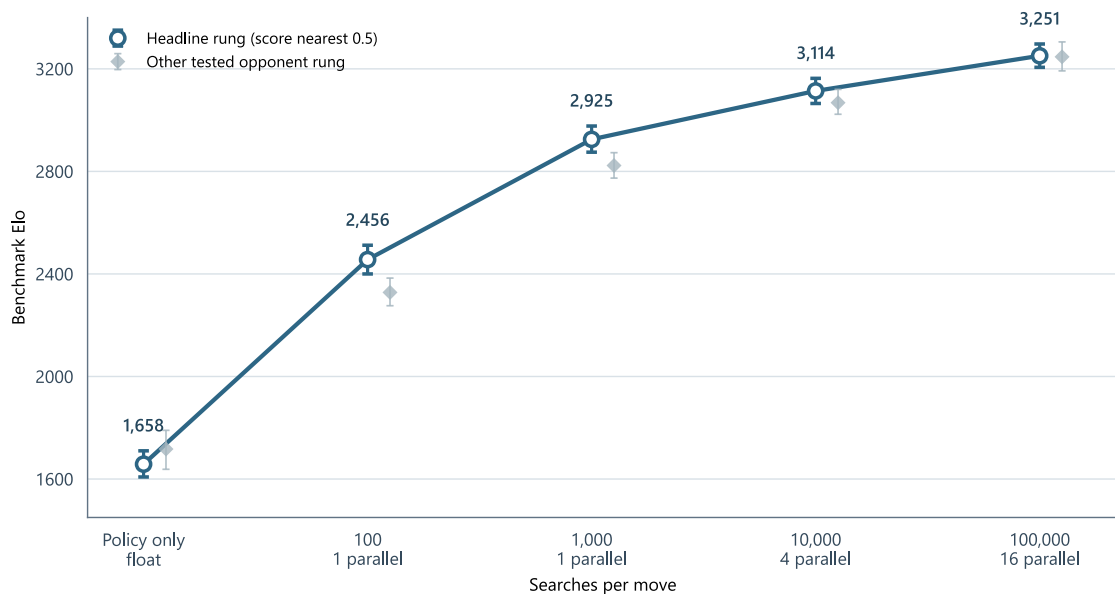


Figure 9: The connected points use the opponent rung with score nearest 0.5; pale diamonds show the other measured rung at each budget. Vertical bars are the reported 95% confidence intervals. The categorical horizontal axis keeps policy-only play visible alongside the searched conditions; it does not imply equal compute spacing. Policy-only uses the float export, while searched points use INT8 TensorRT. The parallel-search count also changes with budget, so this is a measured operating curve rather than an isolated node-budget experiment.

TensorRT, so this is not an architecture-only comparison. Appendix B gives the W/D/L counts and the skipped-rung boundary. Parameter compression is 13.4x; Elo itself is not a meaningful percentage scale.

9 What the evidence does not establish

The complete run establishes the strength of the assembled system, not an Elo contribution for every retained ingredient. Several component comparisons used one seed, a short horizon, frozen replay, or a local proxy. They establish mechanics and screened choices, but not long-run online gains. Historical policy-head and global-context comparisons without complete artifacts remain explicitly qualitative. The larger network reached parity during a limited continuation; that does not distinguish optimization difficulty from a capacity or target-quality limit.

The final ratings come from 100-game paired matches against fixed-node Stockfish 13 and a historical node-to-Elo calibration. They are benchmark Elo, not FIDE ratings or current unrestricted-engine rankings. Match sampling and anchor calibration both matter; the displayed bootstrap intervals include only the former. Opponent rungs disagree at shallow budgets. Policy-only and searched play also use different inference artifacts, while deep-search parallelism varies with budget. Figure 9 is thus a measured operating curve, not an isolated search-scaling law.

The rejected methods' economics depend on this workload. Exact graph reuse and inference-cache hits were too rare in diverse chess self-play; adaptive stopping saved simulations but barely shortened training under actor/trainer overlap. A repetitive analysis service or non-overlapped learner might behave differently. The final student was not matched to the teacher's deployment backend at every budget. Finally, the \$43.20 figure covers only 60 hours of accepted-lineage rental

time; discarded work, evaluation, distillation, and total project spending remain outside it. Reproducibility means a recorded recipe, artifacts, match protocol, and statistical agreement—not bitwise-identical future weights across proprietary drivers and TensorRT versions.

10 Conclusion

In 2.5 days on one eight-GPU node, an AlphaZero-style chess system trained from random initialization produced a 6.3-million-parameter model measuring **3,251 benchmark Elo at 100,000 searches per move** against the project's fixed-node Stockfish 13 ladder. Its **1,658 benchmark-Elo policy-only** result and approximately **74-Elo** matched-estimator gain over the previous 64-search training baseline describe different protocols. Neither is a FIDE rating or a claim against unrestricted engines.

The result belongs to the assembled learning loop. It combines compact structured policy prediction, a shared convolutional network, progressive small-to-medium sizing, searched self-play with restart-state and replay selection, and an INT8-capable architecture. Native tree ownership, batched inference, TensorRT, and distributed training made the necessary volume of searched data feasible. Component measurements explain why these choices were retained, but do not assign each an independent Elo contribution.

The negative results sharpen that conclusion. A learned allocator produced better deep-policy fidelity but worse on-line learning. A stopper saved simulations but barely reduced wall-clock cycle time. Exact graph and neural-cache reuse were too sparse for their overhead in this chess workload. More parallel leaves sped service while spending some playing strength, especially at shallow budgets. Three failures exposed the coupling that those local metrics miss: omitting searched endgames poisoned value targets, a mechanically

successful TensorRT refit changed model behavior, and incomparable training losses promoted a much weaker candidate.

The strongest general lesson is to measure the whole loop. For every proposed saving, ask how many additional reliable positions it creates, how quickly the learner absorbs them, and whether the deployed artifact actually plays better. The remaining uncertainty is also clear: the larger model recovered its parent’s strength without surpassing it in the available continuation, and the compact student preserved substantial strength without matching the teacher. Neither outcome closes the question of how much more chess strength the same framework could gain from longer training, different targets, or more capacity.

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A Training diagnostics

The figures in this appendix describe the accepted training lineage through the selected checkpoint. They do not include the reverted INT8 conversion, the invalid capacity-promotion branch, or the later limited continuation of the larger model. A quantum comprises 500 optimizer steps at a configured global batch of 2,048. The selected checkpoint follows 817 contiguous quanta and records 408,500 optimizer steps and 836,608,000 presentations. The published checkpoint number, 1026, is an artifact identifier, not the number of quanta included in this selected-lineage plot.

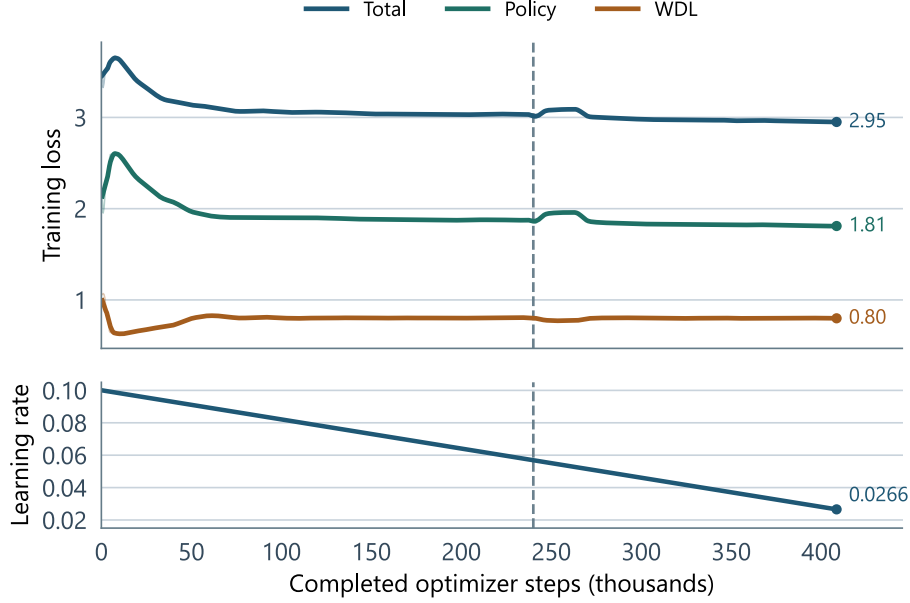


Figure A.1: Total, policy, and WDL training losses are shown as raw traces and 11-quantum moving averages, with final values labeled directly. The dashed line marks the small-to-medium active-model transition at 240,000 optimizer steps. Loss levels across model stages are optimization diagnostics, not paired playing-strength measurements. From 300,000 steps to selection, total loss falls only from about 2.98 to 2.95, while WDL loss stays near 0.80. The active learning rate fell from about 0.10 to 0.0266 by selection.

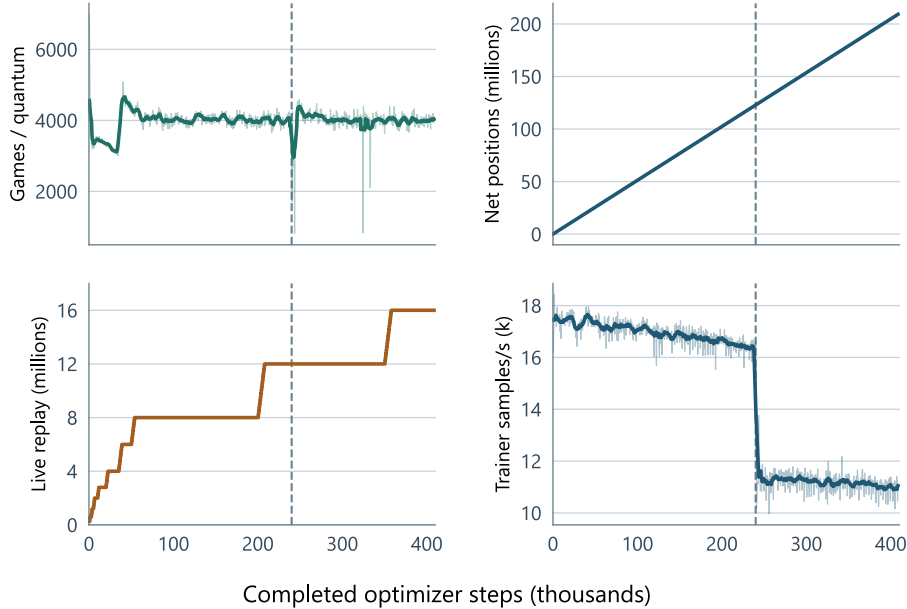


Figure A.2: Four panels separate per-quantum completed games, cumulative net materialized positions, live replay occupancy, and training samples per second on the same optimizer-step axis. The separate replay scale exposes its capacity steps: 209.15 million net positions were materialized, while 16 million rows were live at selection. The small model's median measured trainer throughput was 16,977 samples/s across 480 quanta; the medium model's was 11,194 across 337. Model shape, schedule, and concurrent workload differ, so the gap is not an isolated model-size effect. All panels describe the selected lineage, whose coordinator counters sum to 3,249,647 completed games.

B Evaluation detail

B.1 Fixed-node match protocol

Every teacher row used 100 games from 50 paired openings, once from each colour. Stockfish 13 used one thread and 1,024 MiB hash. Its fixed node limits were assigned the historical benchmark-Elo anchors of 1,700 at 1,000 nodes, 1,890 at 2,000, 2,220 at 5,000, 2,470 at 10,000, 2,700 at 20,000, 2,960 at 50,000, 3,100 at 100,000, and 3,230 at 200,000 [9]. The displayed 95% intervals bootstrap match outcomes while holding these graph-read anchors fixed; they exclude calibration uncertainty. The headline for each candidate budget uses the tested opponent whose observed score is closest to 0.5. Searched games used the INT8 TensorRT artifact; policy-only games used the corresponding float TorchScript export.

For score s against an opponent anchor R , benchmark Elo is $R + 400 \log_{10}(s / (1 - s))$. Each 95% interval resamples the 50 colour-swapped opening pairs 10,000 times, then transforms the 2.5th and 97.5th percentiles of the resulting match scores with the anchor held fixed.

Table B.1: Final checkpoint against fixed-node Stockfish 13

Model searches	Parallel	Opponent nodes	W/D/L	Score	Benchmark Elo (95% CI)
Policy only	–	1,000	32/24/44	0.440	1,658 [1,608, 1,710]
Policy only	–	2,000	16/22/62	0.270	1,717 [1,638, 1,790]
100	1	5,000	51/28/21	0.650	2,328 [2,276, 2,384]
100	1	10,000	39/18/43	0.480	2,456 [2,400, 2,512]
1,000	1	20,000	47/40/13	0.670	2,823 [2,774, 2,873]
1,000	1	50,000	21/48/31	0.450	2,925 [2,875, 2,977]
10,000	4	50,000	45/40/15	0.650	3,068 [3,023, 3,120]
10,000	4	100,000	30/44/26	0.520	3,114 [3,065, 3,163]
100,000	16	100,000	51/38/11	0.700	3,247 [3,192, 3,305]
100,000	16	200,000	25/56/19	0.530	3,251 [3,206, 3,297]

The two deepest-search anchors agree within four Elo; the shallow pairs disagree more substantially. The ladder is an attainable operating curve, not a constant-parallelism search ablation.

B.2 Matched-estimator comparison

The cross-campaign headline does not subtract the displayed 64-search curve endpoints. A retrospective plateau comparison on the same three-rung estimator placed the previous baseline at 2,283.9 and the final recipe at 2,358.0 benchmark Elo, a difference of +74.1. Restricting both plateaus to the clean report-figure windows (3.0 and 2.5 days) gave +75.8. The earlier baseline’s original series used a single-rung fit. A +2.7 Elo estimator transfer, summarized from 35 paired observations in the operator recap, brings it onto the three-rung scale. The underlying paired-observation table is not preserved with the report; approximately ± 15 Elo is a sensitivity allowance for this transfer, not a bootstrap interval. This limitation does not apply to the terminal match intervals above, which come from their game records.

B.3 Distilled student controls

Both students had 470,295 parameters and trained on the same frozen 20-million-row replay snapshot without QAT. They were evaluated with TorchScript and four parallel searches. The 36,621-step student scored 31/33/36 against 20,000 Stockfish nodes at 10,000 searches: 2,683 [2,637, 2,731] benchmark Elo. The 110,000-step student scored 35/29/36 under the same protocol: 2,697 [2,640, 2,753]. At 100,000 searches, the longer student scored 59/28/13 against 20,000 nodes for 2,873 [2,819, 2,935]. That deep point is unbracketed because the planned 50,000-node match was skipped; it is not an architecture-only comparison with the INT8 teacher.

C Supporting search and systems comparisons

The compact controls below preserve denominators that are easy to lose when comparing local speedups. They are measured operating points, not an additive attribution of the final model’s Elo.

C.1 Actor overlap and training supply

On the eight-GPU production-shaped workload, keeping more actors alive during a training quantum traded trainer throughput for ongoing search. The complete-quantum estimate includes the training time and remaining self-play wait; it is not the trainer-only duration.

Table C.1: Actor overlap and training supply

Active actors	Trainer samples/s	Concurrent searches/s	Complete quantum (s)
8	21,492	506,000	117.1
16	17,139	606,000	112.9
32	9,210	742,000	111.2

With no actors active, trainer throughput was 25,275 samples/s, but that excludes a concurrent self-play supply and is therefore not a complete-quantum comparison. Half-active operation was retained near the measured tradeoff. The optimal fraction can change with visit budget, model size, and hardware contention.

C.2 Parallel-search operating point

At 1,000 searches per move against the same 20,000-node Stockfish opponent, parallel leaves substantially shortened the recorded match wall time while reducing measured playing strength. The two timing sources give ranges, not statistical intervals.

Table C.2: Parallel-search strength and wall time

Parallel leaves	Benchmark Elo	Match wall time (min)	Elo relative to serial
1	2,823	18.1–18.5	0
4	2,804	3.4–3.8	−19
16	2,778	1.3–1.8	−45

The relative penalty was much larger at only 100 searches: sixteen parallel leaves cost about 235 Elo. These few budget-by-parallelism points do not define a universal safe concurrency curve. The terminal strength curve varies parallelism with budget and must be read as an operating curve.

C.3 Negative search and reuse controls

The learned pre-search allocator captured about 23% of available deep-policy-divergence headroom at nearly matched spend but trailed non-adaptive online training by roughly 60–100 ladder Elo. A later in-search stopper skipped about 14% of nominal simulations, yet improved generation cadence by only about 3% under actor/trainer overlap. Its paired strength differences ($+1.7 \pm 9.9$ and -4.2 ± 10.1 Elo, standard errors) did not resolve an effect.

Exact graph search avoided only 0.0249% and 0.1769% of neural evaluations at 1,000 and 10,000 searches while being 8.63% and 8.28% slower. A bounded inference cache shared inside each self-play process had a 0.970% hit rate; disabling it made game updates about 0.88% faster and lowered peak worker memory about 7.12%. A wider unbounded repeat tracker observed at most about 4% reuse in the tested production-like workloads and itself cost throughput. These are workload-specific negative findings, not general limits on graph search or caching.

D Reproducibility and release boundary

D.1 Two reproducibility targets

The project maintains two distinct reproducibility targets:

1. **Recipe reproduction:** use the current fully expanded `chess-final-config.yaml` as the supported entry point.
2. **Result reproduction:** use the frozen source revision, resolved config hash, manifest, checkpoints, engines, datasets, and archive recorded for the final result.

The first may evolve; the second must not.

D.2 Result identity and provenance

The frozen local result record identifies these layers of provenance:

- Git source revision and clean/dirty state;
- resolved YAML and SHA-256;
- dependency lock and hash;
- operating image, Python, PyTorch, CUDA, cuDNN, driver, GPU, CPU, RAM, and disk facts;
- Stockfish and KataGo versions and archive hashes where applicable;
- evaluation dataset and opening-suite identities and hashes;
- run manifest, approval record, coordinator logs, TensorBoard events, and resource telemetry;

- replay schema, final capacity/occupancy, and reconciled volume counters;
- selected training checkpoint and trimmed inference artifact hashes;
- ONNX, TensorRT template/engine provenance, calibration positions, and fidelity reports;
- raw terminal match records, aggregate reports, commands, and confidence-interval method;
- one digest covering the fetched archive or a checksummed artifact manifest.

Chapter 8 and Appendix B summarize model identity and evaluation values, not the full provenance hash set. The published INT8 ONNX has SHA-256 `d634abacae3c874eac6ded89f6af861eb81b509da638b5ad710587b1a08be658` at the immutable model-repository revision `dc8fccccb67ab5ec9e36267a165a9700b7dbf55f` [11]. The source release’s evidence index [10] records additional artifact hashes. The frozen training-source revision, resolved configuration, large run archives, and some exact evaluation inputs remain in the local result archive. The public recipe and model support inspection and a new run, but are not yet a self-contained package for exact-match reproduction of the reported experiment.

D.3 Reproducing the software

Local setup and validation begin in the public source-code release [10]. Production nodes are provisioned by `deployment/setup_remote.sh`, which installs locked dependencies, builds the Release extension, installs pinned evaluation engines, and runs engine smokes. Run lifecycle operations go through `deployment/run_control.sh`.

Operational commands are versioned with the source release; they are not part of the scientific match protocol.

Original project code and documentation, including this report, are available under the MIT License in the source release [10]. The published final model artifacts carry the same license in the model repository [11]. External dependencies, reference sources, and third-party data retain their own terms.

D.4 Reproducing evaluation

Use the preserved checkpoint and inference artifact rather than re-exporting it with a newer toolchain. Reuse the same paired opening suite, colors, opponent binary, node limit, threads, hash, candidate search budget, parallelism, batching, and adjudication rules. Report every game and recompute aggregates independently.

For latency, separate:

- isolated model-forward throughput;
- saturated many-position search throughput;
- single-game interactive latency.

Only compare like with like. Appendix B states the terminal opponent, openings, game count, and inference artifact.

D.5 Reproducing plots and tables

The plots derive from archived JSON, CSV, TensorBoard, or manifest data. Figure inputs and extraction methods are retained beside the final evidence archive. Derived tables preserve the raw columns needed to recompute totals, rates, Elo transformations, and uncertainty intervals; live-dashboard values are not treated as publication data.